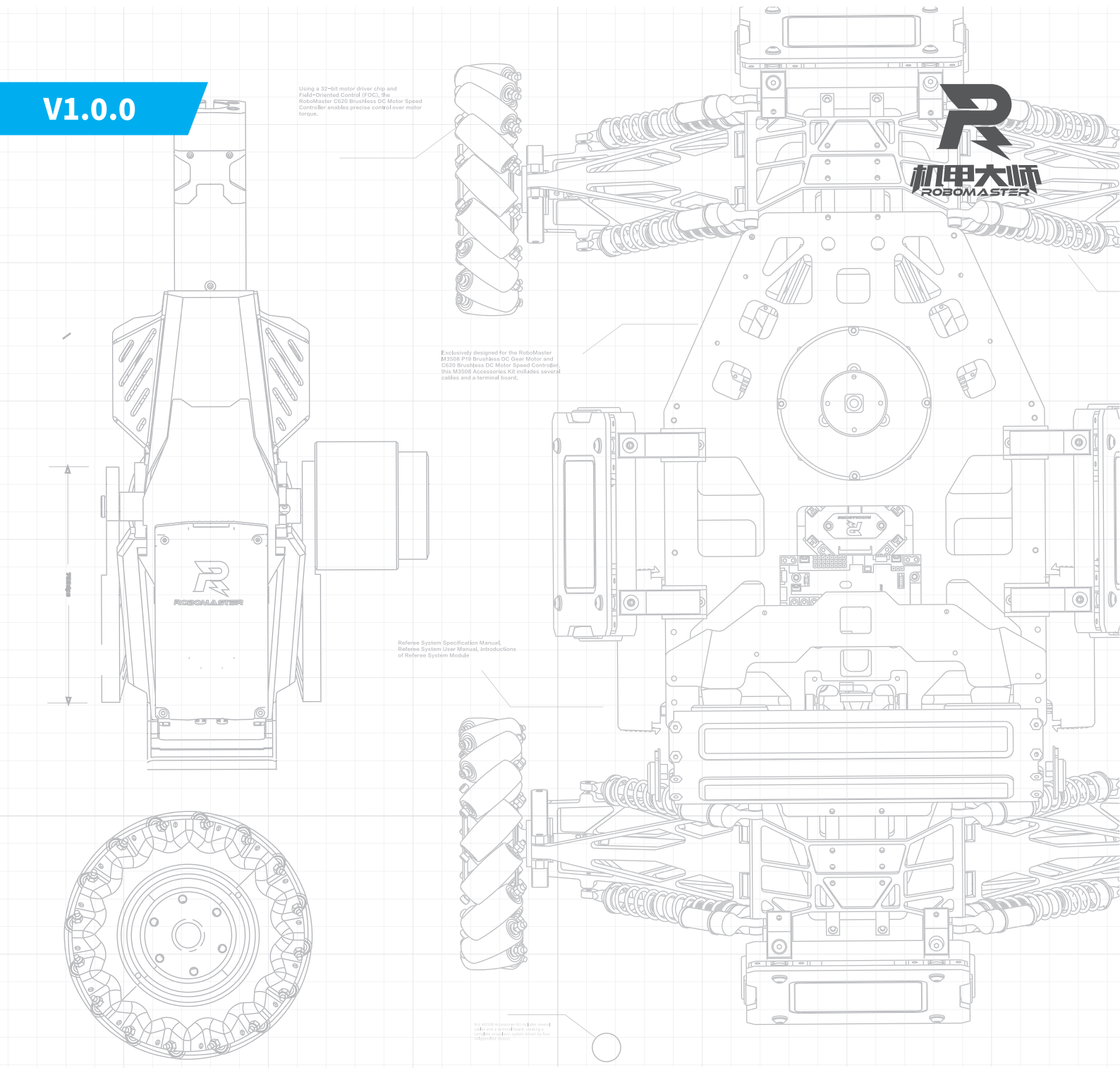
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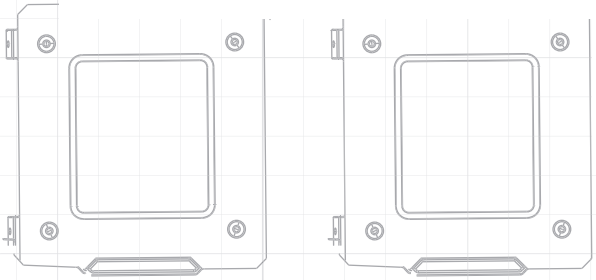


26th National University Student Robot Competition

ROBOMASTER2027

RoboMaster University Series

Robot Building Specifications Manual



Compiled by the RoboMaster Organizing Committee

Published October 2026

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Revision Log

Date	Version	Revision Record
2026.10.15	V1.0.0	First release

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1 Foreword

RoboMaster teams must develop and build their robots themselves. Participating robots must satisfy all specifications described in this document, and Pre-Match Inspection staff will spot-check robots according to its provisions. Robots must also comply with relevant laws and regulations and must not pose safety hazards. If a violation causes a safety accident, RMOC will pursue the violating party's legal liability in accordance with the law. If there is any dispute over these specifications, the interpretation of the Chief Referee or Chief Inspector shall prevail.

Remarks

- Provisions marked “S#” before an item (for example, S10) are mandatory provisions. They focus on key requirements such as safety, size/weight, electromagnetic compatibility, and field rules, and must be strictly observed. For a violation, RMOC may refuse the robot's participation, require rectification by a deadline, or directly disqualify it, depending on the nature of the violation.

- Provisions not marked “S#” are advisory. They are intended to optimize performance, improve stability, or reduce failure risk, but are not hard conditions for passing Pre-Match Inspection. Failure to follow an advisory provision does not directly affect inspection approval, but may cause module malfunction, limited performance during matches, additional repair/damage-assessment costs, or safety hazards. The team bears all resulting losses and consequences.

Unless a provision states otherwise, all mandatory provisions in this manual apply to both RMUC and RMUL robots. RMUL robots may be exempt from a specific requirement if the relevant robot type, field prop, or competition rule does not apply to RMUL.

2 Technical Specifications

2.1 General Technical Specifications

2.1.1 Energy

S1 S1 The only forms of energy a robot may use are electric power and gas supply.

S2 S2 For robots subject to the Chassis Power limit, the power source for horizontal chassis movement is limited to electric power.

- Fuel-driven engines, explosives, and hazardous chemical materials are prohibited.
- Mains power is prohibited within the competition area, except at the Radar computing platform end.
- Hydraulic or other power methods that may cause pollution are prohibited.

2.1.1.1 Power Supply

S3 S3 Batteries must be stably mounted on a battery rack.

Note: The battlefield terrain is complex; it is recommended to provide cushioning protection for the battery.

S4 S4 The only battery types a robot may use are listed in the table below:

Note: For the definition of a regular manufacturer, see “2.3 Definition and Scope of Application of Third-Party Finished Modules and Finished Systems.”

Table 2-1 Summary of Battery Types

Battery type / Robot/Module	Batteries produced by Shenzhen DJI Innovation Technology Co., Ltd. batteries	Produced by other regular manufacturers dry batteries	Produced by other regular manufacturers lithium batteries
Heavy Robot	Allowed	Allowed	Not allowed
Infantry Robot	Allowed	Allowed	Not allowed
Aerial Robot	Allowed	Allowed	Allowed
Hangar	Allowed	Allowed	Not allowed
Sentry Robot	Allowed	Allowed	Not allowed

Battery type / Robot/Module	Batteries produced by Shenzhen DJI Innovation Technology Co., Ltd. batteries	Produced by other regular manufacturers dry batteries	Produced by other regular manufacturers lithium batteries
Dart Launcher	Allowed	Allowed	Not allowed
Dart	Allowed	Allowed	Allowed
Radar sensor end	Allowed	Allowed	Allowed
Radar computing platform end	Allowed	Allowed	Allowed
Custom Terminal	Allowed	Allowed	Allowed

The following provisions in this subsection concerning Supercapacitor module requirements apply only to Supercapacitor modules that participate in Chassis Power.

S5 S5 Only Ground Robots may install Supercapacitor modules; each robot may install at most one Supercapacitor module.

S6 S6 For a Ground Robot, the total nominal energy of a single Supercapacitor module must not exceed 2000 J, and its total measured energy must not exceed 2200 J. The nominal energy of a single capacitor module is calculated as $E = \frac{1}{2} \times C \times U^2$ (where U is the capacitor's rated withstand voltage and C is its capacitance).

S7 S7 For the chassis power source of robots subject to the power limit, the sum of the nominal capacitances of capacitors connected to the Chassis interface but not participating in Chassis Power must not exceed 10 mF.

Note: If the capacitor bank in a Supercapacitor module does not participate in Chassis Power, bring the corresponding schematic to Pre-Match Inspection as supporting evidence.

S8 S8 In a modular 42mm launching mechanism, the sum of the nominal capacitances of the capacitors must not exceed 80 mF.

2.1.1.2 Gas Supply

S9 S9 The pressure of compressed gas stored in a gas cylinder must not exceed 20 MPa, the cylinder's nominal withstand pressure must be at least 30 MPa, and a dual-gauge constant-pressure valve must be installed directly at the cylinder outlet, with a working pressure not exceeding 0.8 MPa.

S10 The working gas must be non-combustible, non-toxic, and non-polluting, such as air, nitrogen, or carbon dioxide.

S11 Any mechanism that uses or stores gas (gas cylinders, gas cylinders/actuators, air springs, etc.) must have a certificate of conformity or stamped nameplate. The certificate or stamp must be easy to view during Pre-Match Inspection.

S12 If a gas cylinder is still within its service life, it must be returned to the factory for maintenance by the deadline specified in its manual or nameplate, and proof of maintenance must be provided.

S13 Gas cylinders must have a certification certificate issued by a certification body officially recognized by the country of origin.

S14 Gas cylinders must be securely mounted to the robot body, with at least two fixing points spaced more than one-fifth of the cylinder's length apart, or a fixing surface longer than one-fifth of its length. Protect the cylinder and gas lines against damage from robot rollover, collision, rotation, or failure of moving parts. The cylinder outlet must not be exposed, to prevent projectile impacts from damaging it. For safety, the outlet must face horizontally or upward.

S15 When a gas cylinder is installed, it must be considered that if the robot rolls over at any angle, neither the cylinder nor the gas lines will touch the ground.

S16 Gas lines and fittings must be suitable for the system's maximum possible working pressure.



- A safety relief valve is recommended for low-pressure gas lines.
- Keep all flammable materials at least 50 mm from the gas cylinder.

2.1.2 Communication Equipment

The competition site and surrounding environment are complex, and RMOC cannot guarantee that the wireless environment in the competition area will not interfere with participating equipment. Teams must adapt to the wireless environment at the competition site.

S17 The designated control methods for this season are listed in the table below:

Table 2-2 Summary of Control Methods

Data link	Information transmitted
Remote Controller	Remote Controller stick information
Camera Video Transmitter Module (receiver end)	Keyboard and mouse commands, Remote Controller stick information
Video transmission link	Custom Terminal information
Referee System serial port	Multi-robot communication



- RMOC recommends that teams use the video-transmission link or the Camera Video Transmitter Module (Receiver) to control robots.
- The Camera Video Transmitter Module (Receiver) itself can transmit remote-control data (and, when paired with the Operator End, can also transmit keyboard and mouse data).
  For the protocols related to the Camera Video Transmitter Module (Receiver), see the Camera Video Transmitter Module User Manual.
- The video-transmission link means the link that uses the Camera Video Transmitter Module for data transmission. For its protocols, see
  RoboMaster 2027 University Series Communication Protocol.

S18 At Pre-Match Inspection, the robot must be equipped with a compliant Remote Controller for debugging, so that the team can demonstrate projectile launching, chassis movement, gimbal movement, and other functions.

S19 The Remote Controllers designated for this season are shown in the table below. Other models of the same style must not be used. Only Remote Controllers using the 2.4G band are allowed, and RMOC does not guarantee their stability.

Note: The Camera Video Transmitter Module (receiver end) has Remote Controller functions but is not regarded as a Remote Controller. During a match, RMOC only guarantees the stability of the Camera Video Transmitter Module (receiver end) located in the Operator Room.

Table 2-3 Summary of Remote Controllers

Name	Picture (for reference only)
DT7	
FlySky i6X	
WFLY ET08, ET08A	

Name	Picture (for reference only)
Radiolink ET16S, Radiolink ET16	

- Given the complex wireless environment at the competition site, these Remote Controllers may also be subject to interference. Using multiple redundant control methods is recommended to avoid loss of control.
- Robots entering the battlefield automatically connect to the Camera Video Transmitter Module (receiver end) in the Operator Room; Camera Video Transmitter Modules (receiver end) brought by teams cannot be used. The sticks on the Operator Room module default to centered, and its switches default to position C.

S20 Except for Aerial Robots, each Remote Controller must be paired one-to-one with a receiver.

S21 Modifying the wireless RF section of a Remote Controller is prohibited. Adjusting how the antenna is fixed, a damaged antenna, or antenna repairs through an official after-sales channel are not considered modifications.

S22 Except for visible light communication, a robot must not carry wireless communication equipment other than the Remote Controller, the Referee System, and radar information receiving devices. Visible light communication is limited to communication between participating robots.

S23 Equipment carried on a robot must not generate radio interference in the 0.428–0.438 GHz, 2.3–2.55 GHz, 3.75–4.75 GHz, or 5–7.5 GHz bands.

Recommendation: Enable all robot functions and measure devices that may generate radio interference with a spectrum analyzer placed close to them.

2.1.3 Optical Means

S24 Except for the Custom Terminal, a robot may carry a display no larger than 7 inches. Illuminance measured within 40 mm of the display must not exceed 300 lux, and the display's emitting surface must be at least 100 mm from the Armor Module.

S25 Except for equipment required by this manual, a robot must not carry obvious visible-light or invisible-light emitting devices.

Light sources used as robot status indicators and infrared sensors are not considered obvious light-emitting devices.

S26 Laser products must not be modified. The safety standard for a laser product is determined using the following data, taking the highest value as controlling:

- The safety label on the laser product, or relevant documentation issued by the manufacturer/distributor.
- The measured power obtained when Pre-Match Inspection staff place a laser power meter directly against the laser product.

S27 A robot must not cause bodily injury to any person through the use of any optical means.

S28 Robots may only install invisible laser products with a Class I laser safety rating. Visible laser products are prohibited.

2.1.4 Visual Features

The competition site and surrounding environment are complex. RMOC cannot guarantee that visual features at the competition site will not cause visual interference. Vision algorithms should adapt to changes in field lighting and other possible interference from the surroundings.

Robot design must comply with the following requirements:

S29 Simulating visual features is prohibited.

S30 Changing a robot's visual features by any means is prohibited.



- In the 2027 season, the hit surfaces of Referee System Armor Modules are dot-matrix screens that can display the corresponding robot number or pattern (the specific style will be updated later).
- Robot visual features: the displayed content and appearance of Armor Modules.
- Field visual features: the Dart Launching Station, visual labels, Energy Mechanism lighting, Technology Core lighting, Dart guidance lights, etc.

2.1.5 Appearance Design

To prevent robot appearance from affecting shooting engagements on the field and the spectator experience, robot appearance design and construction must comply with the following requirements:

Basic requirements:

S31 Robot wiring must be neat, with no exposed metal cores. Any wiring that cannot be concealed must be protected with cable carriers, cable organizers, or similar materials.

S32 A robot's appearance must not include materials that are obviously unsightly, such as washbasins, plastic bottles, corrugated cardboard, bed sheets, white foam board, or air-cushion film.

S33 Fishing nets must not be used as appearance-design materials, but may be used to make Aerial Robot guards.

S34 Sharp structures must not be designed or used, to prevent damage to the field and injury to people.

S35 For Ground Robots, all exposed structural edges 0–400 mm above the ground that may directly contact another robot's Armor Module detection surface must be rounded, with a radius of at least 10 mm.

Flexible protective material will be available for pickup in the Pre-Match Inspection area. However, after application, flexible material may not fit the robot structure, moving mechanisms, or functional components. Teams are therefore advised to install guards or wrap the relevant structures with flexible material before the competition.

Gloss:

S36 The gloss of any robot appearance area larger than 10 mm × 20 mm must not exceed 30 Gs. Items that cannot function if their gloss is reduced, such as optical axes and camera lenses, are exempt, but the shortest straight-line distance from such items to the left or right edge of an Armor Module must exceed 100 mm.

Paint color:

- Recommendation: Keep the appearance styles of all robots on the team consistent.
- Recommendation: Each robot's appearance should include two of its school or team emblems, placed on different sides. Each emblem must be no larger than 100 mm × 100 mm, placed in a prominent location, and at least 30 mm from the light strips on both sides of an Armor Module. If the appearance does not meet the requirements, Pre-Match Inspection staff will require the emblem's position or size to be corrected.

S37 Red Robots may use red-toned protective shells, and Blue Robots may use blue-toned shells. A robot may have at most four areas in the opposing team's color, and each area must not exceed 60 mm × 60 mm.

When designing robots, avoid red- or blue-toned accessories as much as possible.

S38 Printed or stickered school and team emblems must not affect the robot's visual features and must not emit light.

Protective shell installation:

Recommendation: Use tough, damage-resistant materials for protective shells and test their reliability to prevent damage during competition and resulting rule violations.

Decoration requirements:

S39 Advertising areas must be placed on the left and right sides of the robot and at least 30 mm from the edges of Armor Modules.

S40 Printed or stickered advertising areas must not affect the robot's visual features and must not emit light.

S41 Each advertising area on a robot must not exceed 100 mm × 100 mm, and each robot may have at most two areas for sponsor exposure. If the appearance does not meet the requirements, Pre-Match Inspection staff will require the advertising area's position or size to be corrected.

S42 Neither the hardware nor software (including the user interface, UI) of an auxiliary device (Custom Terminal) or its components may display sponsor brand names, logos, slogans, or related elements in any form.

School or team emblems and advertising areas may be rendered in reverse white or retain their original colors.

2.1.6 Launching Mechanism

Launching mechanism: A mechanism that causes a projectile to leave the robot along a fixed path with a certain initial speed.

S43 When compressed gas is used to launch a projectile, the acceleration stroke must not exceed 200 mm (acceleration must be completed before the projectile enters the Speed Monitor Module).

S44 A robot's launching mechanism must be capable of reliably launching 10 17mm projectiles or 5 42mm projectiles.

S45 Each launching mechanism must have the corresponding Speed Monitor Module installed in accordance with the requirements.

2.1.7 Custom Terminal

A Custom Terminal is a multipurpose device built by the team to control robot actions or monitor robot movement and status. It receives match-related information and robot video from the Referee System Game Engine Server and sends competition-specific interactive commands to that server.

S46 The Custom Terminal must meet the following building parameters:

Table 2-4 Custom Terminal Build Parameters

Item	Limit	Remarks
Maximum working voltage (V)	30	The voltage at any point inside the module at any time relative to the battery negative terminal must not exceed 30V
Maximum stowed dimensions (mm, L × W × H)	600*600*600	-
Maximum extended dimensions (mm, L × W × H)	Unlimited	Consequences caused by size, such as inconvenient use or inability to compete normally, are borne by the participating team
Data transmission	Video feed information: UDP Match-related data: MQTT + ProtoBuf	-
Protocol	See the Communication Protocol	-

Installation requirements:

S47 Double-sided tape, screws, or other adhesive or sharp objects must not contact the Operator Room desktop.

S48 No live wires on the Custom Terminal may be directly exposed.

Usage requirements:

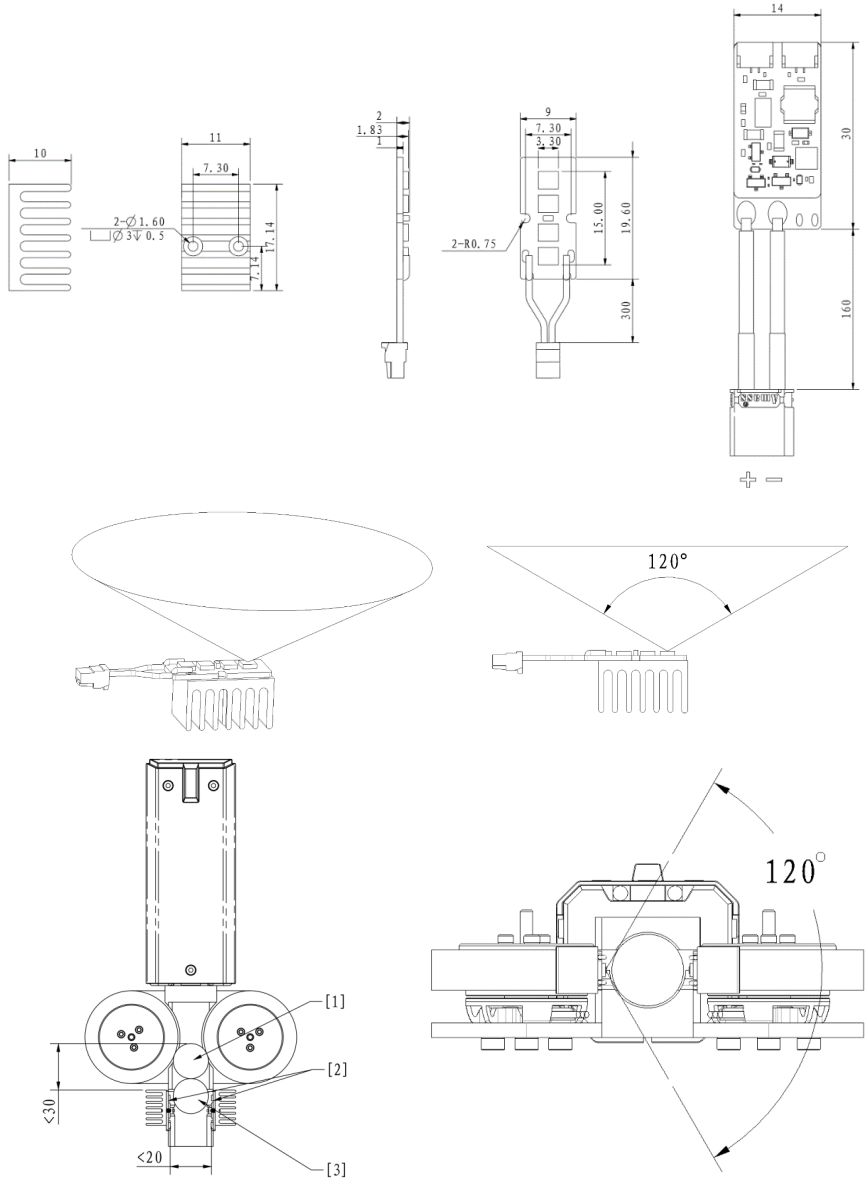
An RJ45 port is provided in the Operator Room for connecting the Custom Terminal signal. Operators may connect it themselves.

S49 Before use, connect the Custom Terminal to the data interface provided in the Operator Room and test that it works properly.

S50 The Custom Terminal must not use wireless transmitters or receivers.

2.1.8 17mm Fluorescent Projectile Charging Device

Refer to the dimensions of the 17mm Fluorescent Projectile Charging Device and reserve mounting holes at a specific position on the robot.



[1] Projectile in the launching position

[2] 17mm Fluorescent Projectile Charging Device

[3] Projectile in the preset position

Figure 2-1 Installation Diagram of the 17mm Fluorescent Projectile Charging Device

The projectile in the launching position is the projectile shown above at the instant it contacts the launching structure, such as the friction wheels. The projectile in the preset position is the projectile immediately before the projectile in the launching position in the feeding path.

Installation steps:

In RMUL, robots do not need to install the Fluorescent Projectile Charging Device.

1. Install the UV light boards at a specific position on the robot. They must be parallel and symmetrically placed about the center plane of the projectile channel. The distance between the upper surfaces of the LEDs on the two boards must be less than 20 mm; the front end of each board must be less than 30 mm from the front of the projectile in the launching position; and each LED must have a 120° illumination FOV. See the figure above for the installation position.
2. If the installed feeding tube is metal, maximize its contact area with the light board and tighten the screws to improve heat transfer. A nonmetal feeding tube must have a heatsink installed.
3. After wiring the UV light board, connect the XT30 port to a 12 V or 24 V power supply as needed.

Installation requirements:

- Recommendation: Install the LED side of the UV light board tightly against the metal part of the launcher and the back of the board tightly against the heatsink. Leave at least 20 mm of clearance outside the heatsink so moving air can contact it directly.
- Recommendation: Provide clearance between conductive materials and the light board power-wire solder pads to prevent short circuits.
- Recommendation: Do not cover the back of the UV light board or the heatsink surface with tape, plastic, or other materials that impair heat dissipation.
- During use, check for dirt obstructing the LEDs or damaged LEDs. Replace them promptly if necessary to avoid affecting Pre-Match Inspection.

S51 Install the UV light board so there is no obvious light leakage, to prevent UV radiation from injuring people.

S52 Align the UV LEDs with the direction of projectile travel in the feeding path. No structure may block the LEDs from illuminating the projectiles, to ensure effective charging.

S53 The 17mm launching mechanism's maximum firing rate must be at least 10 Hz, and the initial projectile speed must exceed 20 m/s. For self-testing, fire at the actual match-condition firing rate or the mechanism's maximum rate, and ensure the brightness meets the self-test requirements. These requirements will be updated later.

S54 During Pre-Match Inspection of the 17mm Fluorescent Projectile Charging Device, the robot must fire at the match-condition firing rate or the launching mechanism's maximum firing rate.

S55 If using the official Fluorescent Projectile Charging Device, use its matching heatsink and accessories.

S56 A homemade UV light board must use UV LEDs with a wavelength of 390–410 nm.

Recommendations for homemade UV light boards:

- Recommendation: Use LEDs with a 120° viewing angle.

- Recommendation: The total luminous power of one light board should be 3 W (6 W for two boards).
- Recommendation: Use an aluminum-core or copper-core circuit board.
- For recommended heatsink specifications, LED spacing, total light-board length, and similar details, refer to the Fluorescent Projectile Charging Device.

2.1.9 Others

S57 Do not use materials that are fragile, easily detached, or difficult to clean during robot design and construction, such as feathers or cotton batting. Do not use any glue or adhesive material to connect the robot to the field or field props.



- If the market definition of a material's fragility or tendency to detach is unclear, the team may use it in a match only after ensuring its strength and obtaining permission from the Pre-Match Inspection referee.
- To avoid potential risks from ambiguous material definitions, such as a robot being unable to enter a match, teams are advised not to use materials whose properties are ambiguously defined as above.

S58 A robot must not leave obvious marks on or damage Energy Units when acquiring or assembling them.



Grasping an onboard Referee System module may seriously affect its function; the team bears the consequences.

S59 Except when a Dart Launcher launches a Dart or an Aerial Robot takes off from its Hangar, a robot must not split into sub-robots or multiple subsystems connected by flexible cables, or actively throw or launch its own parts.

S60 Any onboard computing platform that uses an operating system must use an open-source operating system, such as Linux and its derivatives or an RTOS. Closed-source operating systems such as Windows and macOS are prohibited.

2.2 Specifications for Complete Finished Robots and Open-Source Robots



- Only teams that did not successfully participate in an offline competition of the 2026 University Series may field at most one unmodified RoboMaster Competition Robot Self-Assembled Edition A, or at most one RoboMaster AI Robot 2020 Standard Edition modified to meet the new configuration requirements.
- Other teams building robots themselves must not merely modify the above products into robots with a new configuration or directly use important structural parts such as their frame extrusions. They may use only components from those products such as motor couplings, launching mechanisms, and projectile-feeding mechanisms.

In the RoboMaster University Series, RMOC defines ownership of robot intellectual property as belonging exclusively to the students who participated in its design and construction and the universities they represent. Only a university team that owns the configuration's intellectual property and has the relevant students as team members may compete using that configuration. If another team chooses to reference that configuration, it must

redesign its robot to meet at least one sufficient-redesign condition or three non-sufficient-redesign conditions. The robot's motion mechanism (such as a Ground Robot chassis) and functional mechanism (such as an Infantry Robot launching mechanism or Heavy Robot arm) are assessed independently; the total number of conditions met by both must reach the stated threshold. Once the conditions are met, the design is considered a new configuration that the team may use in competition. Redesign examples for each type are not limited to those listed here.

During the competition, RMOC may conduct spot checks. Teams are responsible for showing RMOC mechanical and electrical design drawings and related code files and answering technical questions. RMOC may photograph or record information about a team's robot configuration.

S61 Before Pre-Match Inspection, teams must submit robot photos through the designated channel. The photos must clearly show the robot's main structure; remove any protective shell before photographing the main body.

S62 When redesigning a complete finished robot or an open-source robot, teams must submit a document describing the relevant modifications through the designated channel.

S63 If robots do not meet the redesign conditions relative to one another and do not reference an open-source robot, the intellectual-property owner is deemed to be the team that first used that robot in an official match. As long as that team remains in the RoboMaster University Series, all robots fielded later by other teams must meet the redesign conditions relative to the previously fielded robot.

S64 When redesigning a complete finished robot or referencing another team's configuration or an open-source design, the Dart Launcher, Dart body, Hangar, and Aerial Robot are each assessed as independent units. Each unit that requires redesign must meet at least one sufficient-redesign condition or three non-sufficient-redesign conditions; conditions are not combined across units.

2.2.1 Sufficient Redesign

Sufficient redesign means improving a robot's core mechanism in a way that has broad effects on related systems.

1. Chassis
   - Change in wheel-group configuration, such as changing among mecanum wheels, steering wheels, omni wheels, Ackermann steering wheels, or tracks;
   - Change in the wheel-group transmission configuration, such as moving the drive source from unsprung to sprung mass, or vice versa;
   - Change in suspension configuration, such as changing among double-wishbone, torsion-beam, double-longitudinal-arm, or lift suspension;
   - Change in chassis drive, such as changing between DC and AC, brushed and brushless, geared and direct drive;
   - Change in chassis form, such as changing among frame-type, layered-plate, and unibody chassis.
2. Gimbal
   - Change in the Gimbal's transmission stack, such as changing the inclusion relationships or number of the Yaw, Pitch, and Roll axes;
   - Change in the projectile-feeding principle, such as changing from a simple direct feed path to feeding through the Yaw-axis joint to the launching mechanism;
   - Change in the magazine location, such as changing whether it is fixed to the chassis, Yaw axis, or Pitch axis;
   - Change in the Gimbal motor transmission method, such as changing among direct drive, belts, linkages, and gears (excluding swapping belts, gears, and sprockets for one another).

3. Actuator

   - Change in the topology of mechanisms other than the end effector, such as changing the order, number, or type of revolute and prismatic joints;

   - Improve actuator power efficiency, for example, by reducing the number of power sources with function unchanged, or increasing effective function while keeping or increasing the number of power sources.

4. Dart System

   - Dart body: Change the overall aerodynamic configuration or the method of generating control forces. For example, change from a conventional main-wing-and-tail layout to a flying-wing layout;
     or substantially restructure the function allocation of the front and rear aerodynamic control surfaces, with corresponding redesign of the aerodynamic layout, actuators, and control coordination.

   - Dart Launcher: Change the principle of the launching power or the topology of the core transmission; change the kinematic-chain topology of the core feeding or loading mechanism.
     For example, change from elastic-energy ejection to friction-wheel acceleration; redesign the drive kinematic chain so the principle by which power is input, transmitted, and applied to the Dart changes.

5. Aerial System

   - Hangar: Change the principle or kinematic-chain topology of the core storage, transfer, or docking mechanism. For example, change from independently parking and launching at each bay to a shared launch platform with an active transfer mechanism; or change from fixed passive-guided docking to active multi-axis alignment, with corresponding restructuring of sensing, actuation, and control.
   - Aerial Robot: Change the overall arrangement of lift units or the principle of the attitude-control mechanism; change the main load-bearing structure. For example, change from coplanar rotors to coaxial placement, or from fixed-axis rotors to actively tilting rotors, thereby changing how loads transfer among power units, onboard equipment, and landing structures.

2.2.2 Non-Sufficient Redesign

Non-sufficient redesign means an improvement with limited effects on related systems.

- A change of more than 10% in a key parameter (such as suspension hardpoint position, track width, wheelbase, Dart Launcher effective acceleration stroke, Dart body, or local Aerial Robot aerodynamic geometry);
- Change in the number of effective powered wheel groups;
- Change in chassis, Gimbal, Dart Launcher, or Hangar drive-motor gear ratio, or in an actuator's local transmission path or method;
- Change the placement orientation of at least three onboard Referee System modules, with each change causing a topology change in the robot system;
- Add a new functional module related to the robot's existing functions, such as an independent rescue device, small video-transmission gimbal, Energy Unit orientation mechanism, or vision module; Dart Launcher loading or launching modules; or Aerial System/Dart-body sensing, feedback, or actuation modules.
- Change between longitudinal and transverse friction-wheel placement;

- Change the power source, for example, pneumatic to electric;
- Change the end-effector type or configuration, for example, rotary grasping to translational grasping;
- Change the layout of core electronic components (main controller, power supply, computing platform, sensing modules), for example, move them from the Gimbal to the chassis;
- Change the Gimbal's rotation range from limited to unlimited;
- Adjust the size, number, or mounting angle of local aerodynamic components on the Dart, such as tail area, tail mounting angle, or airfoil camber;
- Change the Dart material and adjust its mass distribution or surface properties;
- Adjust the layout or structure of the Dart Launcher feeding mechanism, such as changing the shape of the feed track or optimizing the projectile pusher shape;
- Revise the transmission ratio or path between Dart Launcher components.

2.2.3 Invalid Redesign

Invalid redesign means a modification that has little effect on core functions.

Invalid redesign includes, but is not limited to, the following modifications:

- Modify non-primary load-bearing structures or cover/guard structures, such as changing perforations or shape;
- Change the quantity or model of standard parts;
- Swap a pneumatic cylinder for an electric pushrod;
- Routine counterweight adjustment, changing only the servo or battery model, or changing only the mounting interface that mates with the launching mechanism;
- Substitute materials with the same characteristics, such as swapping fiberglass and carbon-fiber plates;
- Make a change without changing the principle, such as changing a rigid upper feed tube to a soft tube;
- Change only the Hangar outer case or hatch shape while copying the same parking bays;
- For an Aerial Robot, change only the propellers, motors, battery, or flight-controller model; change only PID or off-the-shelf firmware settings; or add only a non-primary load-bearing propeller guard;
- For a Dart Launcher, change only the motor, spring, or rubber-band specification, or change only the shell or Dart mounting method.

2.3 Definition and Scope of Third-Party Finished Modules and Finished Systems

S65 Unless otherwise specified, teams must not use third-party finished modules or finished systems when building robots.

- Third party: Any entity other than RMOC or the participating team.
- Finished module: A specific functional component composed of several basic functional components that can be used to build a complete system, such as:

Robot arm, chassis, Supercapacitor control board and associated code, Gimbal, Dart, suspension wheel group, projectile-feeding mechanism, launching mechanism, motherboard, CPU, GPU, memory, and hard drive.

- Finished system: A complete functional system composed of several finished modules or a combination of self-built modules (including at least one finished module), such as a laptop or all-in-one desktop computer.



Computing platforms used by the Radar computing platform end and Custom Terminal are exempt from this provision and may use finished modules and finished systems.

Exemption for General-Purpose Commercial Products

To encourage teams to use general-purpose products with lasting value in scientific research and engineering practice, commercial general-purpose products may be used and are not considered violations if they meet all the conditions below.

Commercial general-purpose products must meet all of the following conditions:

- Public-sales requirements
  - Manufactured or sold by a legally registered company or organization;
  - Publicly sold to the general public before the registration deadline through channels such as the company website, e-commerce platforms, or distributors, and not only to RoboMaster teams or through channels for a specific event.
- General-purpose use requirements
  - The product is primarily intended for general industrial, scientific research, educational, or consumer markets;
  - Promotional materials must not present RoboMaster or any single other event as the product's sole or primary use case;
  - A product whose functions or specifications are clearly customized for RoboMaster or another specific event, or which is sold only through event channels, is not considered a commercial general-purpose product.

Price limits

- Finished module: Unit price must not exceed RMB 5,000;
- Finished system: Unit price must not exceed RMB 8,000;
- Price is based on the official list price or the tax-inclusive transaction price on a mainstream sales channel. RMOC reserves the right of final interpretation.

Computing platform exemption conditions

A computing platform must meet the public-sales and general-purpose-use requirements above, and must meet either the price condition or the peak-computing-power condition:

- Price condition: same as for finished systems;
- Peak-computing-power condition:
  - Accelerated computing power must not exceed 160 TOPS (based on the AI TOPS INT8 sparse value);

- General-purpose computing power must not exceed 25 GHz. Calculate it by summing the nominal base clock frequency of every processor core (that is, Σ(frequency × number of cores) ≤ 25 GHz).

Compliance certifications

In addition, the certifications below may help support a determination that a product is a commercial general-purpose product.

- Electronic and electrical products should meet the basic safety and electromagnetic-compatibility standards normally required in their target market;
- For products sold in China, any of the following certifications is strong evidence that the product was designed and manufactured for the general market:
  - China Compulsory Certification (CCC), for products in the mandatory certification catalog;
  - IECEE CB Scheme certification;
  - International or regional safety certifications such as UL, ETL, TUV, or CSA;
  - Electromagnetic compatibility or radio certifications such as FCC, IC, CE, RCM, or KC.

Prohibited circumstances and discretion

Even if some of the conditions above are met, RMOC may still determine that a product is a prohibited finished module if any of the following applies:

- It was clearly developed or customized specifically for RoboMaster or another event, and is sold to teams only in related groups, internal channels, or on a limited basis;
- Although displayed online, its main promotional slogan, product name, or packaging presents "for RoboMaster only," "for use only in [an event]," or similar as a key selling point;
- Its launch is clearly tied to the season cycle, appearing only briefly before or after a season without a sustained sales record for a broad market;
- It attempts to disguise a product clearly associated with an event as a general-purpose product through simple renaming or minor revisions;
- RMOC determines that the product's integration replaces core technical challenges of the competition to such an extent that the competition loses its intended technical challenge and fairness.

RMOC may require the team to provide the following materials to determine the product's status:

- Product purchase invoices or order records;
- Product manuals, datasheets, advertisements, or promotional materials;
- Product website or e-commerce page links and historical versions;
- Relevant certification documents or test reports;
- Other materials proving the product's launch date, sales scope, and purpose.

RMOC reserves the right to make the final interpretation and determination of whether a product is a "commercial general-purpose product."

2.4 Robot Technical Specifications

- Maximum stowed dimensions: The robot must have a configuration that meets the maximum stowed-dimension limit, including the Referee System and its brackets.
- Maximum extended dimensions: The maximum dimensions of the robot at any time, measured with its mechanical structure extended to its limit, including the Referee System and its brackets. When a combined system and a system unit are physically connected, the system unit is included in the combined system's maximum extended dimensions. During Pre-Match Inspection, the team must actively demonstrate all mechanical transformations. The robot must meet the maximum extended-dimension requirements in every form before, during, and after transformation.
- In this section, the robot's length, width, and height correspond to the X, Y, and Z axes of its body coordinate system, respectively. For the Aerial Robot and Radar, the X and Y axes may be chosen arbitrarily, and the Z axis is perpendicular to the supporting plane. The Dart System is not subject to the coordinate-system requirements.

S66 Except for the Radar System, Custom Terminal, and modular 42mm launching mechanism, a robot must not be separated into independent units when its dimensions change. Its connecting structures must meet the following requirements:

- Mechanical hinges or ball joints (such as hinges, pivots, spherical bearings, or linear guides), and other movable connections that interlock through solid-body structures, are permitted.
- Flexible tether connections (such as cables, ropes, chains, or belts) and connections maintained by suction/friction (such as:
  suction cups, magnetic attraction, or connections relying only on friction) are prohibited.

Adjusting the tightness of screws, cable ties, or similar parts without removing them is not considered separation into independent units.

2.4.1 Heavy Robot

S67 The Heavy Robot must be capable of horizontal movement and picking up Energy Units in the Resource Zone.

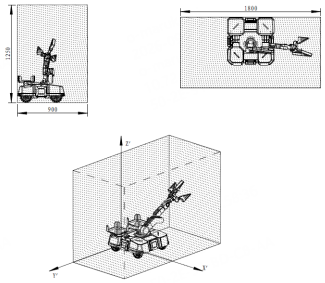
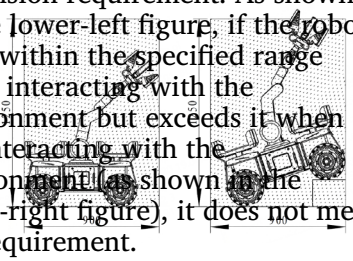
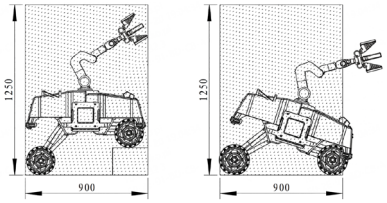
In RMUL, the Heavy Robot does not need to be capable of picking up Energy Units.

S68 The Heavy Robot must meet the following building parameters:

Table 2-5 Heavy Robot build parameter description

Item	Limit	Remarks
Operation mode	Unrestricted; at most 1 Remote Controller and 1 Custom Terminal configured	-
Maximum total power supply capacity	300	-

Item	Limit	Remarks
Maximum operating voltage (V)	30	The voltage at any point inside the robot at any moment must not exceed 30V relative to the battery negative terminal
Launching mechanism	Modular 42mm launching mechanism	Can only be installed during the seven-minute Round
Maximum weight (kg)	- 36: includes battery and Referee System weight, but excludes the modular 42mm launching mechanism weight. - 41: includes battery, Referee System and modular 42mm launching mechanism weight.	-
Maximum stowed size (mm, L × W × H)	The robot has a state with length, width and height not greater than 600 × 600 × 600	The maximum stowed size does not include the modular 42mm launching mechanism
Maximum extended size (mm, L × W × H)	One of the following requirements must be met: • Requirement 1: In any situation, the length, width and height of the robot must not exceed 1200 × 1200 × 1100. • Requirement 2 (both must be met at the same time): (1) Take the outer edge of the body on the positive X axis direction of the robot body coordinate system as the Y' axis, choose any point on the Y' axis as the origin O to establish an X'Y'Z' coordinate system. With the robot chassis locked, other movable structures extending in the negative X axis, positive Y axis and negative Y axis directions must not go beyond x'∈[-900,0], y'∈[-900,900], z'∈[0,1250] (2) Take the outer edge of the body on the negative X axis direction of the robot body coordinate system as the Y' axis, choose any point on the Y' axis as the origin O to establish an X'Y'Z' coordinate system. With the robot chassis locked, other movable structures extending in the positive X axis, positive Y axis and negative Y axis directions must not go beyond	- The modular 42mm launching mechanism must be installed during Pre-Match Inspection. • The actual Pre-Match Inspection method for Requirement 2: place the robot on level ground with the robot against the limit surface and the positive X axis direction of the robot perpendicular to the limit surface. The robot may be translated along the limit surface to select its initial position. After the initial position is determined, the chassis must not translate or rotate, and with the robot's mechanisms in any reachable combination of states, no part of the robot may go beyond the cuboid space with the floor as the ground, the rear side as the limit surface, a forward depth of 900mm, a lateral width of 1800mm (900mm on each side of the lateral centerline) and a vertical height of 1250mm. Example: As shown in the left figure below, the robot in this posture meets the maximum extended size requirement. Afterwards, without any deformation,

Item	Limit	Remarks
	The robot must extend to $x' \in [0, 900]$, $y' \in [-900, 900]$, $z' \in [0, 1250]$. 	…the requirement. After that, if the robot does not transform, an extension beyond the specified range caused by interacting with the environment (such as crossing terrain, as shown in the lower-right figure) is not considered a failure to meet the maximum extended-dimension requirement. As shown in the lower-left figure, if the robot stays within the specified range while interacting with the environment but exceeds it when not interacting with the environment (as shown in the lower-right figure), it does not meet the requirement.  
Referee System	- Small Armor Module • Video Transmitter Module with camera (transmitter end) • Field Interaction Module • Positioning Module (not applicable in RMUL) • Main Controller Module • Power Management Module • Light Indicator Module	-

S69 The modular 42mm launching mechanism must be capable of launching 42mm projectiles.

S70 The modular 42mm launching mechanism must meet the following building parameters:

Table 2-6 Modular 42mm launching mechanism build parameter description

Item	Limit	Remarks
Maximum total power supply capacity (Wh)	-	No battery may be installed; it is powered directly by the robot
Maximum operating voltage (V)	140	The voltage at any point inside the module at any moment must not exceed 140V relative to the battery negative terminal
Maximum weight (kg)	6	Includes Referee System weight
Maximum stowed size (mm, L × W × H)	The modular 42mm launching mechanism has a state with length, width and height not greater than 400 × 400 × 400	-
Referee System	42mm Speed Monitor Module	

The modular 42mm launching mechanism will be disassembled and reassembled multiple times during inspection. Teams are advised to keep each disassembly/reassembly to under 1 minute; teams bear the consequences if they delay the inspection process.

2.4.2 Infantry Robot

S71 The Infantry Robot must be capable of horizontal movement and launching 17mm projectiles.

S72 The Infantry Robot must meet the following building parameters:

Table 2-7 Infantry Robot build parameter description

Item	Limit	Remarks
Operation mode	Unrestricted; at most 1 Remote Controller and 1 Custom Terminal configured	-
Maximum total power supply capacity (Wh)	300	-
Maximum operating voltage (V)	30	The voltage at any point inside the robot at any moment must not exceed 30V relative to the battery negative terminal

Item	Limit	Remarks
Launching mechanism	1 17mm launching mechanism	-
Maximum weight (kg)	28	Includes battery weight and Referee System weight
Maximum stowed size (mm)	The robot has a state with length, width and height not greater than 600 × 600 × 600	-
Maximum extended size (mm)	In any situation, the length, width and height of the robot must not exceed 800 × 800 × 800	-
Referee System	- Small Armor Module - 17mm Speed Monitor Module • Video Transmitter Module with camera (transmitter end) • Field Interaction Module • Positioning Module (not applicable in RMUL) • Main Controller Module • Power Management Module • Light Indicator Module - 17mm fluorescent projectile charging device • Supercapacitor Management Module	The weight is 3.25kg

2.4.3 Sentry Robot

S73 The Sentry Robot must be capable of horizontal movement and launching 17mm projectiles.

S74 The Sentry Robot must meet the following building parameters:

Table 2-8 Sentry Robot build parameter description

Item	Limit	Remarks
Operation mode	Unrestricted; at most 1 Remote Controller used for debugging	-
Maximum total power supply capacity (Wh)	300	-

Item	Limit	Remarks
Maximum operating voltage (V)	30	The voltage at any point inside the robot at any moment must not exceed 30V relative to the battery negative terminal
Launching mechanism	1 17mm launching mechanism	-
Maximum weight (kg)	28	Includes battery weight and Referee System weight
Maximum stowed size (mm)	The robot has a state with length, width and height not greater than 600 × 600 × 600	-
Maximum extended size (mm)	In any situation, the length, width and height of the robot must not exceed 800 × 800 × 800	-
Referee System	- Small Armor Module - 17mm Speed Monitor Module • Field Interaction Module • Positioning Module (not applicable in RMUL) • Main Controller Module • Power Management Module • Light Indicator Module - 17mm fluorescent projectile charging device • Supercapacitor Management Module • Video Transmitter Module with camera (transmitter end)	The weight is 3.25kg

2.4.4 Aerial System

The Aerial System consists of an Aerial Robot and a Hangar.

S75 The Aerial Robot must be capable of hovering continuously for at least 10 seconds.

S76 The Aerial Robot must meet the following building parameters:

Table 2-9 Aerial Robot build parameter description

Item	Limit	Remarks
Operation mode	Fully automatic, with a Remote Controller for emergency propeller shutdown	Can communicate with the Hangar while inside the Hangar
Max total power supply capacity (Wh)	4	-
Max operating voltage (V)	30	At any moment, the voltage at any point inside the robot must not exceed 30V relative to the battery negative terminal.
Launching mechanism	Installation of a launching mechanism is not allowed	-
Max weight (kg)	x	Size and weight limits are linear: $y = (0.3225 - x) / 0.001225$, where $x \in [0.15, 0.249]$. Round the result to the nearest integer. When $x < 0.15$, $y = 141$.
Max extended size (mm)	When the Aerial Robot weighs x, its length, width, and height must not exceed $y \times y \times y$ under any circumstances.	

S77 The team must provide a rigid, open-top container (such as a box without a lid) sized to fit each Aerial Robot. During Pre-Match Inspection, the robot's measured weight is used to calculate its maximum permitted extended dimensions, and the robot is placed in the container. The container's internal length, width, and height must not exceed the permitted limits for that weight, and the robot must remain within the interior space in every extended state, as shown in "Figure 2-2 Aerial Robot Pre-Match Inspection Diagram."

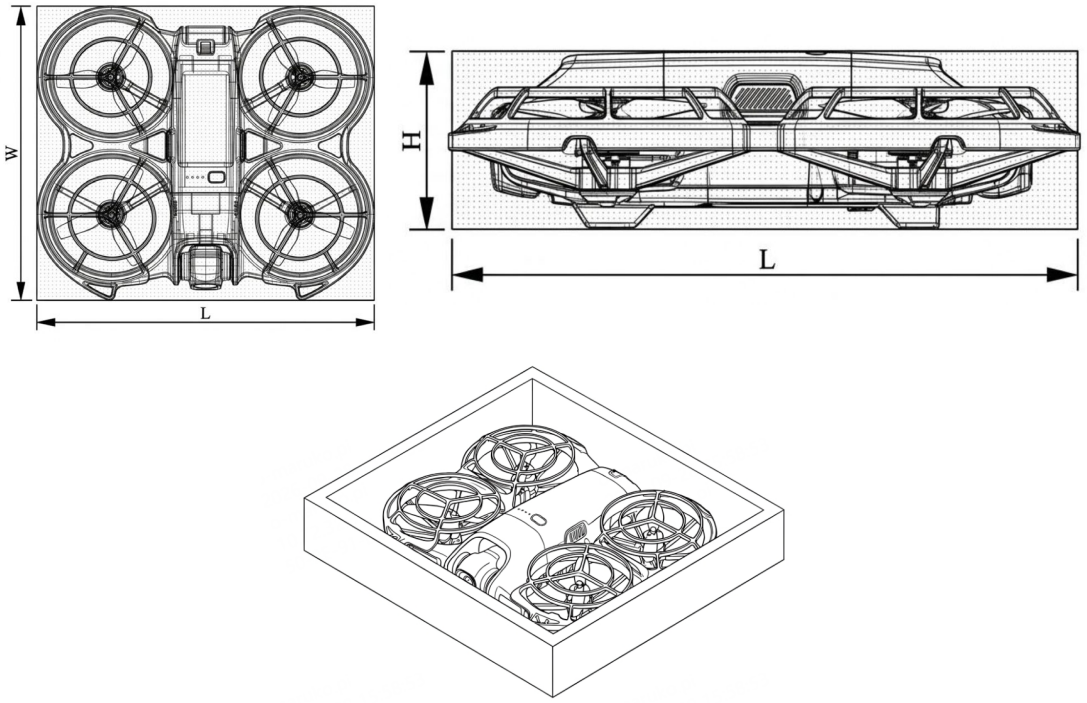


Figure 2-2 Aerial Robot Pre-Match Inspection diagram

S78 All Aerial Robot propellers must have omnidirectional protective structures. These structures must fully enclose each propeller's swept volume, forming a continuous closed boundary above, below, and around the propeller. The openings in the protective structure must be smaller than 17 mm. During operation, the structure must not break, detach, or deform into the propeller's swept volume.

The following two tests are required during Pre-Match Inspection:

1. A 17mm projectile placed at any point on the protective structure must not pass through it;

2. Drop the Aerial Robot freely from a height of 500 mm onto a floor covered with field mat, once each from the bottom, side, and top. The protective structure and propellers must not show obvious deformation or damage.

Flexible material is recommended for Aerial Robot propellers.

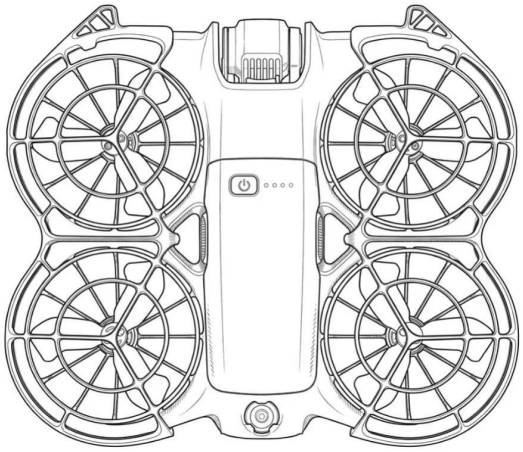


Figure 2-3 Aerial Robot propeller guard diagram

S79 The Aerial Robot must have a Remote Controller for emergency propeller shutdown and may pair only with that controller. At any time, this controller must be able to shut down the propellers of all Aerial Robots on the field simultaneously. In addition, if the Aerial Robot loses the Remote Controller signal at any time, its propellers must stop automatically.

- During Pre-Match Inspection, turn off the Remote Controller to demonstrate the Aerial Robot's automatic propeller-stop function.
- After its propellers stop, an Aerial Robot may fall onto the field and be hit or run over by other robots. It may also experience a large impact when striking a target. Teams are advised to design for strength and shock absorption to prevent damage.

S80 The team must design and install appearance lights to make the robot more recognizable in flight. The lights must ensure that the projection of the Aerial Robot from above can be clearly observed. Requirements follow:

- The appearance lights must use a circular lamp group installed at the center of the Aerial Robot's top surface. It must be securely and naturally fitted to the body, with no obvious tilt, offset, partial obstruction, gaps, bulges, or wrinkles. The angle between the normal of the mounting surface and the positive Z axis of the body coordinate system

must not exceed 10°. The mounting surface must be at the highest point of the body, excluding motors, motor mounts, and propellers.

- The lamp group must use seven XL-3528SURUBC LEDs (red/blue, SMD3528 package, no built-in control IC, front-emitting). The LEDs must be evenly distributed, with one at the center. The diameter of the circle enclosing the outermost edges of the LEDs must not exceed 24 mm.

- The lamp group must use DC drive and remain continuously lit at its rated voltage and current. Periodic PWM or scanning drive is prohibited. Within the specified exposure range of the designated camera, there must be no light-dark stripes or frame-to-frame brightness fluctuation that affects recognition. The LEDs must not have obvious brightness or color differences; their emitted color must be stable, with no obvious color cast or locally white areas.

- The lamp group must switch between red and blue to match the team's side in the match. The Red team's lamp group must show red (target RGB value #FF0000); the Blue team's must show blue (target RGB value #0000FF). Red teams must turn on the red LEDs and turn off the blue LEDs; Blue teams must turn on the blue LEDs and turn off the red LEDs.

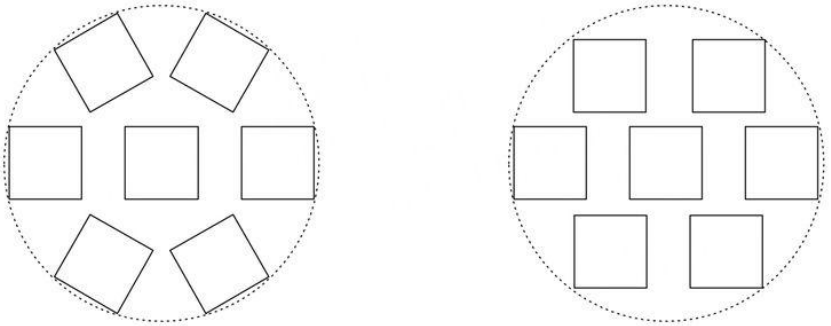


Figure 2-4 Light group diagram

The self-test requirements for the appearance lights will be updated later.

S81 Teams may install indicator lights on the Aerial Robot for communication. Their color must match the team's side in the match: Red teams must show red, and Blue teams must show blue.

S82 The Hangar must be capable of storing the Aerial Robot.

S83 The Hangar must meet the following building parameters:

Table 2-10 Hangar build parameter description

Item	Limit	Remarks
Operation mode	Unrestricted; at most 1 Remote Controller used for debugging	-
Maximum total power supply capacity (Wh)	300	-

Item	Limit	Remarks
Maximum operating voltage (V)	30	The voltage at any point inside the robot at any moment must not exceed 30V relative to the battery negative terminal
Aerial Robot loading capacity limit	4	-
Maximum weight (kg)	Unlimited	-
Maximum stowed size (mm, L × W × H)	300*300*200	-
Maximum extended size (mm, L × W × H)	600*300*400	Refer to the Hangar platform in the "RoboMaster 2027 University Championship Rules Manual"; the Hangar platform is adjacent to the Power Rune pillar, so pay attention to the extension direction
Referee System	- Main Controller Module • Power Management Module • Video Transmitter Module with camera (transmitter end)	-

2.4.5 Dart System

The Dart System consists of a Dart and a Dart Launcher. The Dart Launcher carries the Dart and provides its initial propulsion.

S84 The Dart must meet the following building parameters:

Table 2-11 Dart build parameter description

Item	Limit	Remarks
Maximum total power supply capacity (Wh)	4	-
Maximum operating voltage (V)	8.4	The voltage at any point inside the robot at any moment must not exceed 8.4V relative to the battery negative terminal
Maximum weight (kg)	0.3915	Includes the dart trigger device

Item	Limit	Remarks
Maximum extended size (mm, L × W × H)	250*250*150	- The length along the dart flight direction must not exceed 250 • The dart wingspan must not exceed 250

The forward direction of the Dart Trigger Device is the Dart's flight direction; y and z in the figure are the Dart's wingspan and height, respectively.

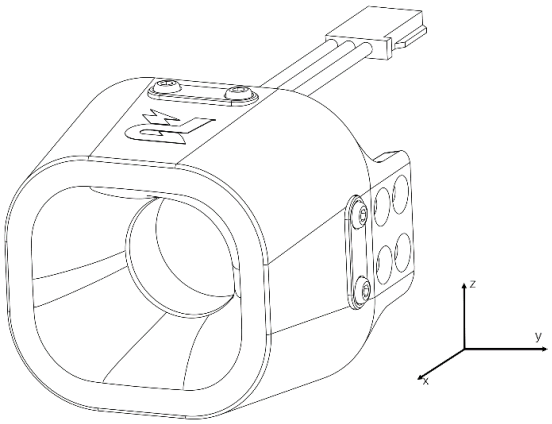


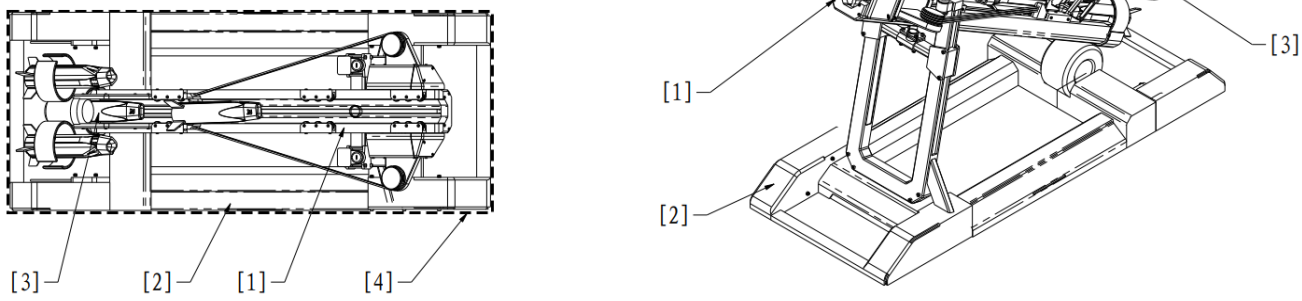
Figure 2-5 Dart flight direction diagram

S85 The Dart Launcher must meet the following building parameters:

Table 2-12 Dart launcher build parameter description

Item	Limit	Remarks
Operation mode	Unrestricted; at most 1 Remote Controller configured	Adapting to the Referee System serial port is recommended
Rotation angle (°)	- Yaw axis angle: unlimited • Pitch axis angle: 25-45	-
Maximum total power supply capacity (Wh)	300	-
Maximum operating voltage (V)	30	The voltage at any point inside the robot at any moment must not exceed 30V relative to the battery negative terminal
Maximum operating power (W)	Unlimited	-
Dart loading capacity limit	4	-
Dart Initial Launch Speed	Unlimited	When the dart Initial Launch Speed exceeds 25m/s, the Referee System

Item	Limit	Remarks
		The Dart may not be detected
Maximum weight (kg)	25	Includes battery and Referee System weight; no dart needs to be loaded when measuring the maximum weight
Maximum extended size (mm, L × W × H)	1000*600*1000	- The maximum extended size is the size with the Dart launcher loaded with 4 darts • In any situation, the orthographic projection of the Dart System on the ground must not go beyond a 1000 × 600 rectangular area, and the highest point of the Dart System must be no more than 1000 above the ground • The orthographic projections on the ground of the Dart launcher and the dart to be launched must at no time go beyond the outermost edge of the Dart System bottom frame. The part of the outermost edge of the Dart System bottom frame that contacts the ground must be a closed rectangle of size 950 × 550, with an assembly tolerance of ±50mm allowed, as shown in "Figure 2-6 Dart System size limit diagram".
Referee System	- Main Controller Module • Power Management Module	The weight is 0.22kg



[1] Dart Launcher
[2] Bottom of the Dart System
[3] Dart body
[4] Outermost edge of the contact plane

Figure 2-6 Dart System size limit diagram

S86 The Dart Launcher must be capable of loading and launching a Dart more than 5 m, and the Dart must have a Dart Trigger Device installed. During Pre-Match Inspection, the team must provide video showing this Dart launching system launching a Dart more than 5 m.

S87 The Dart must target the Dart Detection Module and control its flight direction by means such as a propeller (at most one), control surfaces, or jets, ultimately striking the Dart Detection Module to score a hit. The Dart's thrust-to-weight ratio must be less than 1.

S88 The Dart Trigger Device must light up when the Dart is launched. From a top-down view, no exposed light source other than the Referee System and Dart Trigger Device may be visible on the Dart System.

If the Dart Trigger Device does not light up when the Dart is launched, the Referee System cannot properly detect the launch.

S89 The Dart Trigger Device has built-in red and blue LEDs and must be set to the color corresponding to the team's side in the match.

- After launch, the Dart may land on the field and be hit or run over by other robots. It may also experience a large impact when hitting a target. Teams are advised to design for strength and shock absorption to prevent damage.
- If the Dart Trigger Device shows abnormal behavior, such as alternating red and blue flashes or damaged LEDs, it is damaged, and the Base and Outpost may not detect a Dart hit. Replace it with a spare device; the team bears all resulting losses if it does not.

S90 Directly using compressed air to propel a Dart is prohibited.

S91 The Dart System may be in a ready-to-launch state only during the seven-minute Round.

Ready-to-launch state: A stored-energy component that provides the Dart's initial kinetic energy is tensioned, pressurized, or rotating. Examples include, but are not limited to, rubber bands, gas cylinders, and friction wheels.

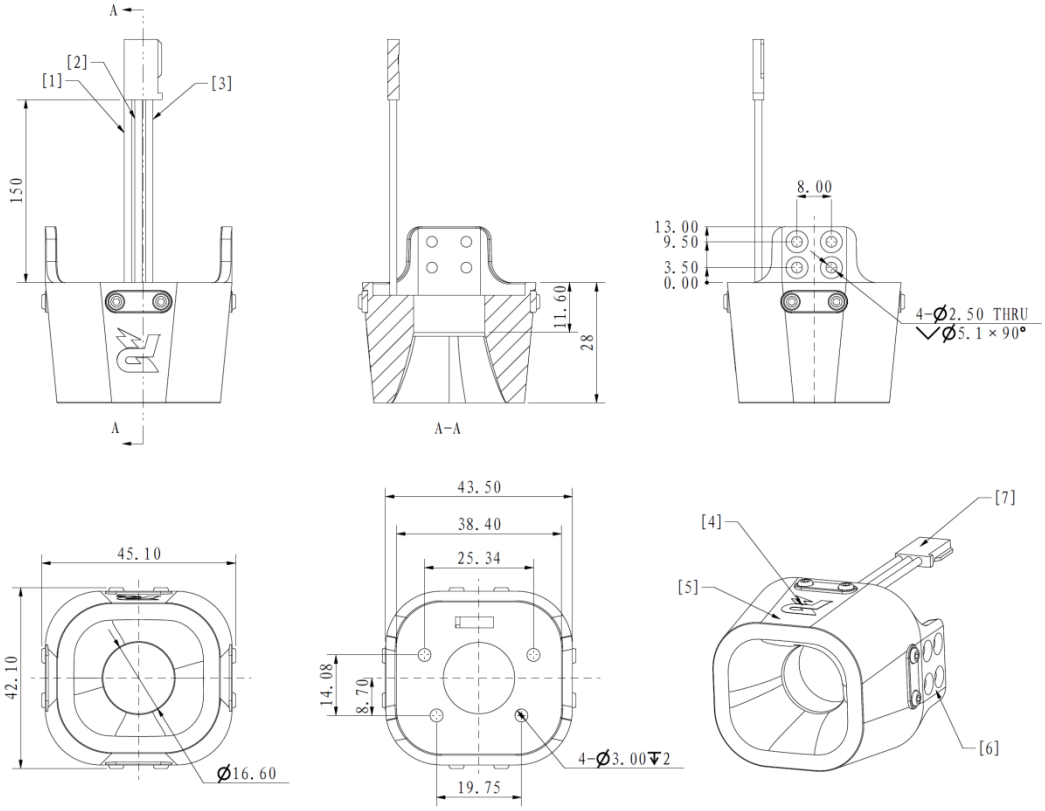
2.4.5.1 Installation Specifications

S92 The Dart must have the officially designated Dart Trigger Device installed.

The officially designated Dart Trigger Device for this season is the "Dart Trigger Device 2."

The Dart Trigger Device weighs approximately 39.5 g ± 2 g; its appearance and dimensions are shown below.

Refer to the Dart Trigger Device dimensions and reserve mounting holes at the front of the Dart.



[1] Power negative	[2] Power positive	[3] Serial cable	[4] Logo
[5] Dart Trigger Device	[6] Mounting lug	[7] 3P 2.54 mm ribbon cable	

Figure 2-7 Dart trigger device diagram

2.4.5.1.1 Installation Steps

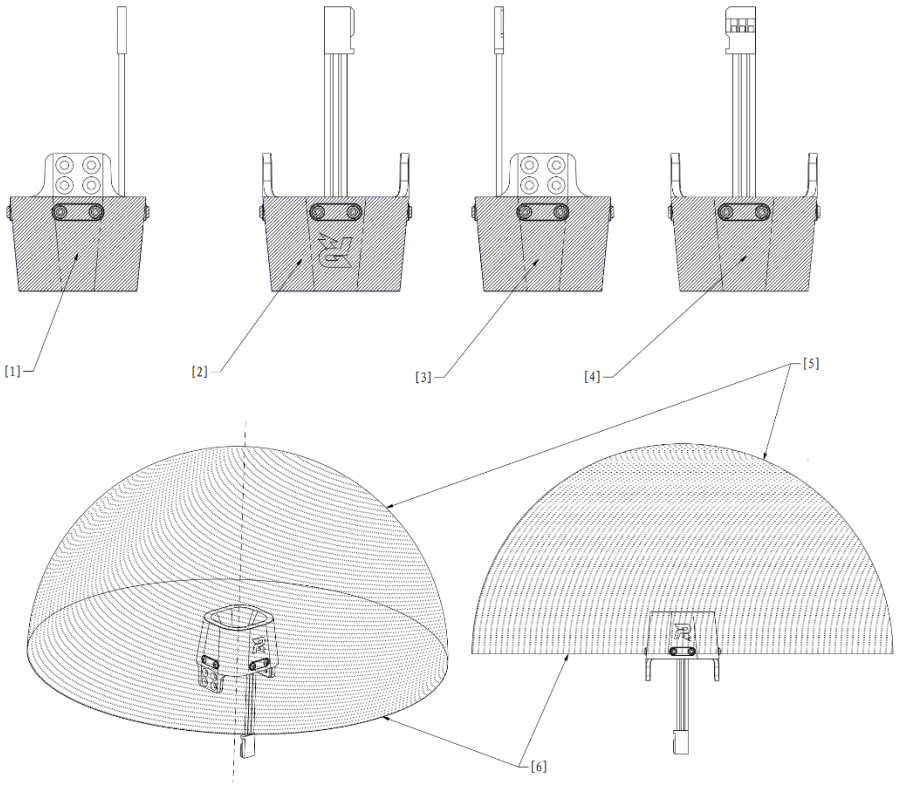
1. Use eight M2.5 countersunk screws and four locating pins to secure the Dart Trigger Device to the front of the Dart.

2. Connect the Dart Trigger Device power port to a 5 V supply.

2.4.5.1.2 Installation Requirements

S93 The Dart Trigger Device may become dim when battery voltage is low. During a match, the battery voltage used by the Dart must remain at least 4 V. If the device's actual operating voltage is below 4 V, the Base and Outpost may not detect a Dart hit.

S94 After installation, ensure that the Dart structure does not obstruct the top, bottom, left, or right of the Dart Trigger Device. No obstruction is allowed within the upper hemisphere of 125 mm radius centered on the center of the device's bottom surface, as shown below.

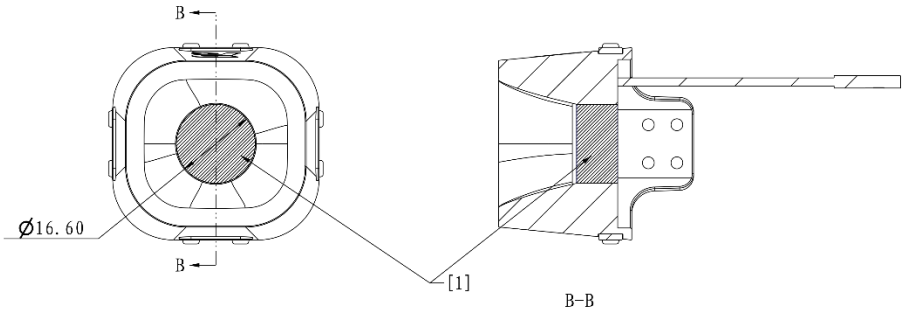


[1] Left [2] Above [3] Right

[4] Below [5] No-obstruction zone [6] Unrestricted area inside Dart Trigger Device

Figure 2-8 Dart trigger device no-obstruction zone diagram

S95 A Dart camera or other device may be installed inside the cavity of the Dart Trigger Device, but the installation area must not extend beyond the shaded region shown below.



[1] Shaded region

Figure 2-9 Dart trigger device internal cavity diagram

2.4.5.2 Visual Features

The visual feature used to assist Dart System aiming is the integrated green LED on the Dart Detection Module. For its specific dimensions, see

the description of the Outpost and Base in the RoboMaster 2027 RoboMaster University Championship Rules Manual.

2.4.6 Radar System

S96 The Radar System consists of a computing platform end and a sensor end. They must be connected by cable and be able to display a debugging view.

S97 The Radar computing platform end must meet the following building parameters:

Table 2-13 Radar computing platform end manufacturing parameters

Item	Limit	Remarks
Operating mode	Fully automatic; at most 1 Remote Controller configured for debugging	-
Max power (W)	750	If the total power generated during Radar System use exceeds the max power, mains power to the Radar computing platform end will be cut off for that round, and all resulting consequences are borne by the team
Supply voltage (V)	220	This is the mainland China mains standard; other countries and regions may refer to the local mains standard, or use another power supply that meets general power supply standards
Supply frequency (Hz)	50	This is the mainland China mains standard; other countries and regions may refer to the local mains standard
Max extended size (mm)	Under any circumstances, the length, width and height of the Radar computing platform must not exceed 600 × 350 × 600	-
Referee System	- Main Controller Module • Power Management Module	Weight 0.22 kg

S98 The Radar sensor end parameters are as follows:

Table 2-14 Radar sensor end parameters

Item	Limit	Remarks
Max extended size (mm)	The length and width of the Radar sensor end must not exceed	- The Radar sensor bracket height is recommended to be 1.2 m

Item	Limit	Remarks
	Radar sensor-end platform height Must not exceed 1500 under any circumstances	and above • Connecting cables are excluded from the dimensions • The radio receiver covered by the Radar sensor end may be separate from its main body, but its length and width must not exceed those of the platform supporting the Radar sensor end
Max operating voltage (V)	60	The voltage at any point inside the robot must not exceed 60 V relative to the battery negative at any time

2.4.6.1 Installation Specifications

The Radar sensor end and computing platform end are installed far apart. Teams are advised to provide at least 5 m of usable cable length.

2.4.6.2 Computing Platform End

During the three-minute Setup Period, the team must place the computing platform end on the designated table near the Radar Base. The Referee System for the Radar may be installed only on the computing platform end.

S99 Wireless devices must not be installed or used on the computing platform end. If a wireless device cannot be removed, it must be disabled in the operating system.

S100 The Main Controller Module and Power Management Module must be installed securely on the Radar computing platform end in accordance with the module installation specifications. The Referee System may share the computing platform end's power supply or use a battery designated for this season.



- Official AC power is 220 V, 50 Hz, from one five-hole outlet under the new national standard. The single-phase two-pole and single-phase three-pole sockets may be used simultaneously. Teams must prepare a power adapter as needed.
- RMOC guarantees only that the official display at the Radar Base and the HDMI cable provided by RMOC will work properly. Teams must resolve any connection issues themselves.

2.4.6.3 Sensor End



Teams must install protective covers on their equipment to prevent projectile impacts from damaging it during a match.

S101 The sensor end must be secured to the Radar sensor bracket and placed on the Radar Base.

S102 The wireless receiver must be installed on the Radar sensor end.

S103 Teams must design their own Radar sensor bracket to raise the sensor installation height.

S104 The Radar sensor bracket must be sized to fit on the mounting surface of the Radar Base. Refer to the description of the Radar Base in the RoboMaster 2027 RoboMaster University Championship Rules Manual for its specifications. The team is responsible for signal transmission and power for the sensor end.

3 Referee System Installation Specifications

Users are advised to first read the manuals for the Referee System modules to understand their functions and installation. After installing each module correctly, read the Referee System User Manual to understand the system as a whole.

Download: https://bbs.robomaster.com/wiki/20204847/804050?source=7

When installing and using the Referee System, follow these recommendations:

- During robot operation, no Referee System module should collide or rub against the ground, walls, or other objects in a way that may cause damage.

- Connect Referee System modules in the manner specified in their manuals and provide suitable protection to ensure stable connections.

S105 Except for the installation methods described in this manual, disassembly or modification of any Referee System onboard end is prohibited.

S106 A robot may install only the Referee System borrowed for the current season.

S107 When replacing Referee System onboard-end modules during a match or Pre-Match Inspection, observe the following time requirements (excluding firmware-upgrade time):

Table 3-1 Maximum replacement time for each Referee System onboard end module

Module	Max replacement time (minutes)
Main Controller Module	5
Power Management Module	10
Light Indicator Module	5
Small Armor Module	2
Armor support frame	7
Camera Video Transmitter Module (transmitter end)	5
Field Interaction Module	7
17mm Speed Monitor Module	7
42mm Speed Monitor Module	7
Positioning Module	4

Module	Max replacement time (minutes)
Supercapacitor Management Module	3

S108 If a robot is noncompliant with the building specifications while overturned, the referee may waive a penalty based on the circumstances. However, a state that the robot deliberately enters is not considered an abnormal operating condition.

3.1 Referee System Overview

Teams must design their robots with mechanical and electrical interfaces reserved and comply with this chapter when correctly installing each Referee System module.



- RMOC provides Referee Systems on loan. Teams may obtain loan eligibility through the technical reviews for each event. See “Season Schedule” in the relevant event participant manual for details.
- Chassis Power: The power consumed by mechanisms that use rotational motion and ground friction to move or rotate a robot horizontally on a plane. For example, steering motors on steering wheels, joint motors on legged robots, and track mechanisms used only to climb steps or cross obstacles are not counted toward Chassis Power.
- A team’s equipment must not interfere with proper operation of the Referee System. If interference occurs, the team must cooperate with RMOC to adjust the source until it no longer causes interference.
- When a robot passes through the Tunnel, the Speed Monitor Module and Positioning Module may be temporarily obstructed. The Main Controller Module, Camera Video Transmitter Module (Transmitter), and Light Indicator Module may also be temporarily obstructed, but the team bears responsibility for resulting malfunctions or interference. All other modules must continue to meet their respective installation requirements.

The Referee System consists of the following modules:

Table 3-2 Referee System modules

Module	Description
Main Controller Module	The Main Controller Module is the core control unit of the Referee System. It can monitor the running status of the whole system and integrates functions such as human-machine interaction, wireless communication and status display.
Power Management Module	The Power Management Module controls power to the robot's chassis, gimbal and launching mechanism, forwards data, detects Chassis Power, and so on.
Light Indicator Module	The Light Indicator Module uses LED light bars to indicate the robot's red/blue side, HP, Buffs, module offline status and so on.
Armor Module	The Armor Module detects projectile hits and collisions on the robot.

Module	Description
Speed Monitor Module	The Speed Monitor Module detects the robot's projectile Initial Launch Speed and Barrel Heat. It is divided into the 17mm Speed Monitor Module and the 42mm Speed Monitor Module.
Field Interaction Module	The Field Interaction Module can exchange information with the field or with the Field Interaction Module cards of robots to implement the corresponding functions.
Camera Video Transmitter Module	The Camera Video Transmitter Module is divided into a transmitter end and a receiver end. The transmitter end is installed on the robot, and the receiver end is installed on the client in the Operator Room. It captures the view in front of the robot in real time through the camera and sends the first-person view back to the display in the Operator Room. It also provides remote control of the robot.
Positioning Module	The Positioning Module obtains the robot's position on the Battlefield.
Supercapacitor Management Module	The Supercapacitor Management Module detects the energy input to and output from the supercapacitor module.

3.2 Robot Referee System Onboard End Configuration

The onboard Referee System module configurations for each robot are listed below:

Table 3-3 Robot Referee System onboard end module configuration

Robot type / Quantity	Heavy Robot	Infantry Robot	Aerial System	Sentry Robot	Dart System	Radar System
Main Controller Module	1	1	1	1	1	1
Power Management Module	1	1	1	1	1	1
Light Indicator Module	1	1	0	1	0	0
Small Armor Module	4	4	0	4	0	0
Camera Video Transmitter Module (transmitter end)	1	1	1	1	0	0
Field Interaction Module	1	1	0	1	0	0
17mm Speed Monitor Module	0	1	0	1	0	0

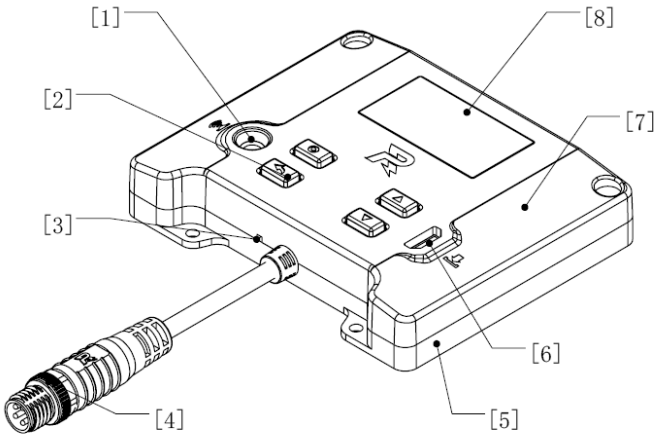
Robot type / Quantity	Heavy Robot	Infantry Robot	Aerial System	Sentry Robot	Dart System	Radar System
42mm Speed Monitor Module	1	0	0	0	0	0
Positioning Module	1	1	0	1	0	0
Supercapacitor Management Module	1	1	0	1	0	0



In RMUL, robots do not need to install a Positioning Module.

3.3 Main Controller Module Installation Specifications

Refer to the Main Controller Module dimensions and reserve mounting holes at a specific position on the robot.



[1] Infrared receiver	[2] Button	[3] Power indicator	[4] Aviation cable connector with black metal ring
[5] Metal bottom shell	[6] Upgrade port	[7] Plastic top shell	[8] Screen

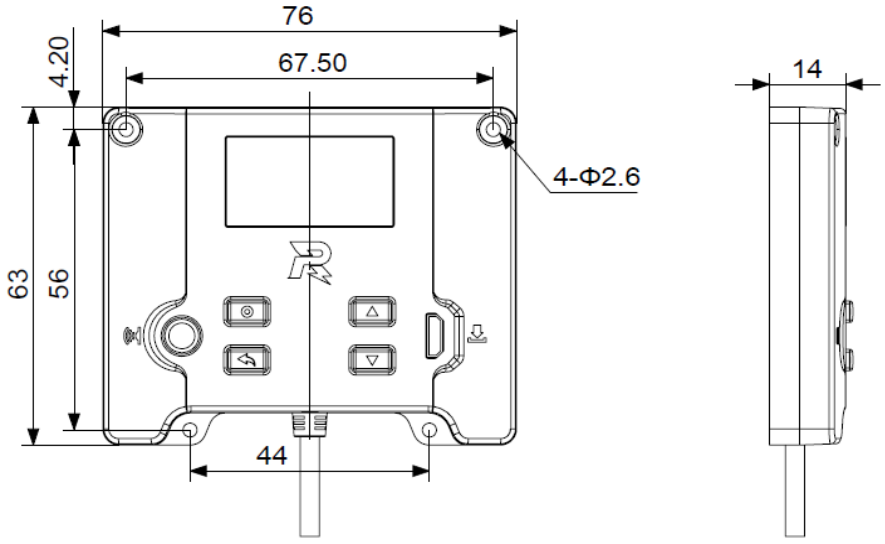


Figure 3-1 Main Controller Module diagram

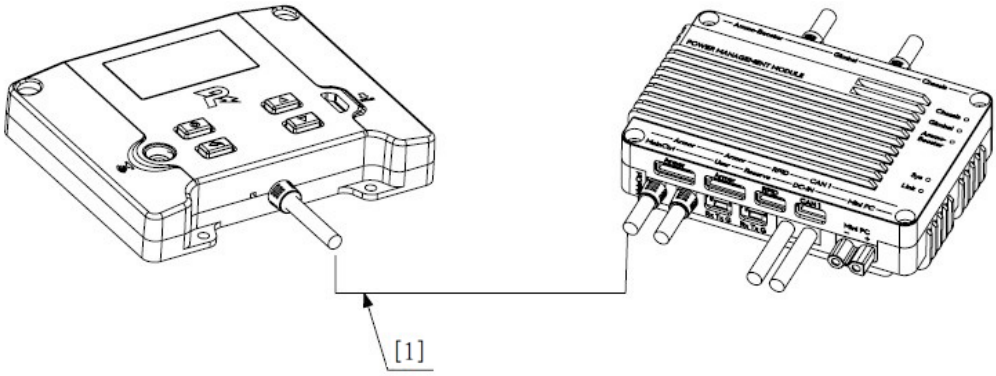
3.3.1 Installation Steps

1. Use four M2.5 screws to secure the Main Controller Module at the designated position on the robot.

Installation reference: You may design your own parts and protect all sides with nonmetallic panels to prevent projectile impacts.

2. Use the aviation adapter cable included in the package to connect the aviation cable connectors on the Main Controller Module and the Power Management Module (both have black metal rings).

Do not connect any other Referee System onboard-end module in series between the Main Controller Module and Power Management Module.



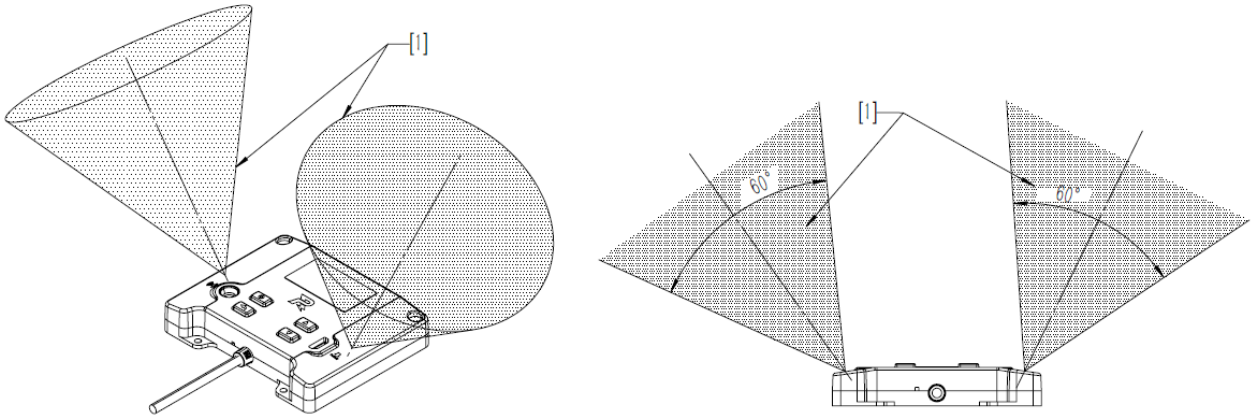
[1] Aviation adapter cable

Figure 3-2 Main Controller Module wiring diagram

3.3.2 Installation Requirements

Install the Main Controller Module in accordance with the following requirements; otherwise, connection to the Game Engine Server may fail or be unstable.

- When the robot is operating, the top surface of the Main Controller Module must face horizontally upward. Using the center of each antenna as the vertex, there must be at least one unobstructed cone with a 60° cone angle (solid angle $\pi(2-\sqrt{3})$ sr) extending infinitely, or an equivalent space that does not interfere with wireless signal transmission. The equivalent space must be one continuous cone, and its full cone generatrices after adjustment must lie in the upper hemisphere. See the figure below.

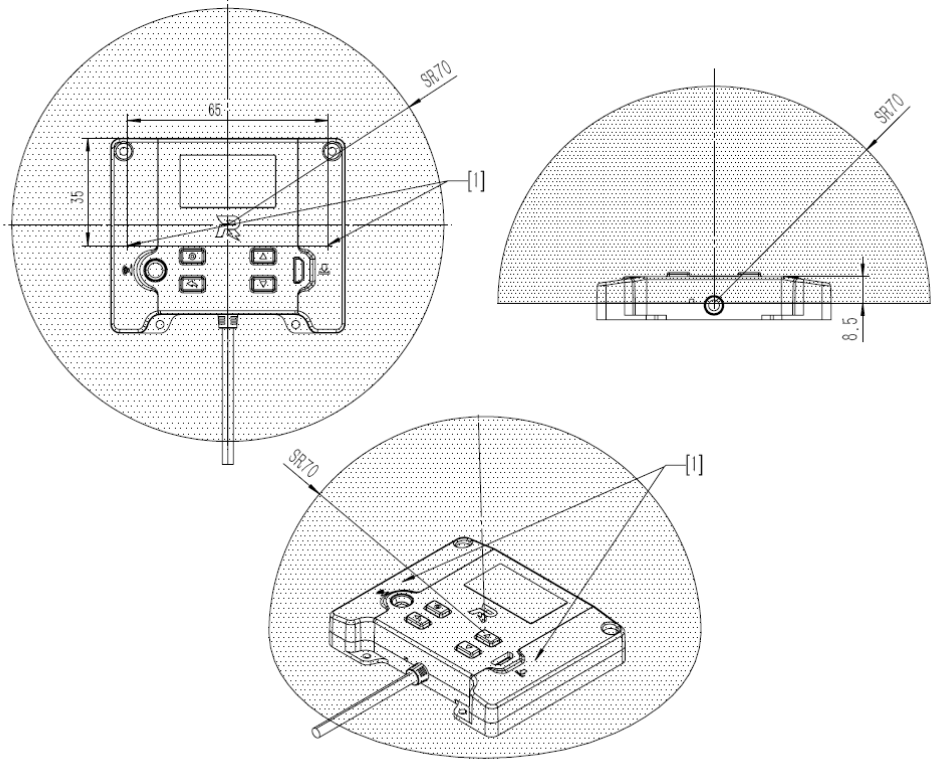


[1] No-obstruction zone

Figure 3-3 Main Controller Module no-obstruction zone

- Install the Main Controller Module in a well-ventilated location; otherwise, overheating may cause it to malfunction during a match.
- Avoid obstructing the screen, buttons, ports, or infrared receiver on the Main Controller Module. Any protective cover must be transparent and easy to open. If it cannot be opened, the screen, buttons, ports, and infrared receiver must be fully exposed. Obstruction may make connection to the Game Engine Server impossible and prevent troubleshooting.
- Within the upper hemisphere of 70 mm radius centered 8.5 mm directly below the center of the logo, avoid conductive materials, electromagnetic-absorbing materials, or other components that may interfere with Wi-Fi signals.

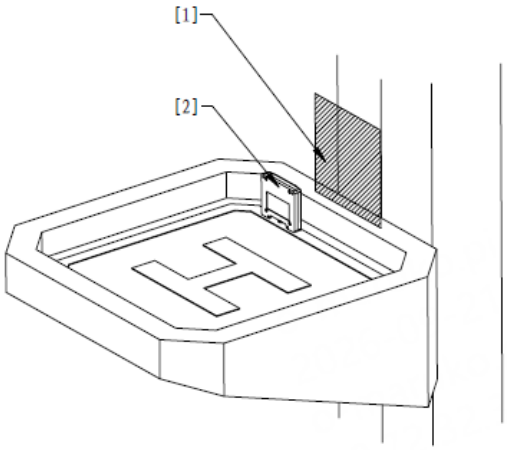
If electromagnetic shielding or interference-generating equipment carried by a Gimbal, robotic arm, or similar mechanism moves through this area, it may interfere with the Wi-Fi signal.



[1] Antenna center

Figure 3-4 Main Controller Module installation position diagram

S109 A 150 mm long and 150 mm high infrared-emitting area is provided on the Energy Mechanism column. It is centered across the column's width, with its lower edge flush with the bottom surface of the Hangar platform. Teams must use this area's location to design the Main Controller Module's installation position and orientation in the Hangar, ensuring that its infrared receiver faces the area and is within its effective reception range so it can connect to the server.



[1] Infrared-emitting area [2] Main Controller Module

Figure 3-5 Hangar Main Controller Module installation diagram

If the device does not connect to the server within a certain time after power-on during the Setup Period, its position may be adjusted within the infrared range.

S110 During official matches, the Dart System and Radar System must connect the Power Management Module and Main Controller Module using the method designated by RMOC.

For the Radar System:

The battlefield provides a wired connection for the Radar System to connect to the battlefield network and server. The Radar computing platform end has a wired connection point: an aviation cable with a single female connector. Field staff must connect the Main Controller interface on the Radar computing platform end's Power Management Module to this wired connection point using the official aviation docking cable.

To connect quickly and successfully, follow these requirements:

- The Radar System does not need to carry its own Main Controller Module during a match. Remove the debugging Main Controller Module from the robot before inspection.
- The Radar System Power Management Module must be exposed and placed where staff can easily access it.
- Do not connect any device to the Main Controller interface on the Radar System Power Management Module. Arrange and store this aviation cable neatly, without tangling, so it can be pulled out and connected easily to the wired connection point.

For the Dart System:

The battlefield provides a wired connection for the Dart System to connect to the battlefield network and server. The Dart Launching Station provides an 800 mm aviation cable with a single female connector. Field staff must connect the Dart System to the wired connection point using the official aviation docking cable.

To connect quickly and successfully, follow these requirements:

- The Dart System does not need to carry its own Main Controller Module during a match. Remove the debugging Main Controller Module from the robot before inspection.

- The aviation cable connecting the Power Management Module and Main Controller Module must be easy to plug and unplug without tools.
- The Dart System Power Management Module must be exposed and placed where staff can easily access it.
- Install the Dart System Power Management Module as close as possible to the wired connection point at the Dart Launching Station; do not connect anything to its other end. Arrange and store this aviation cable neatly, without tangling, so it can be pulled out and connected easily to the wired connection point.

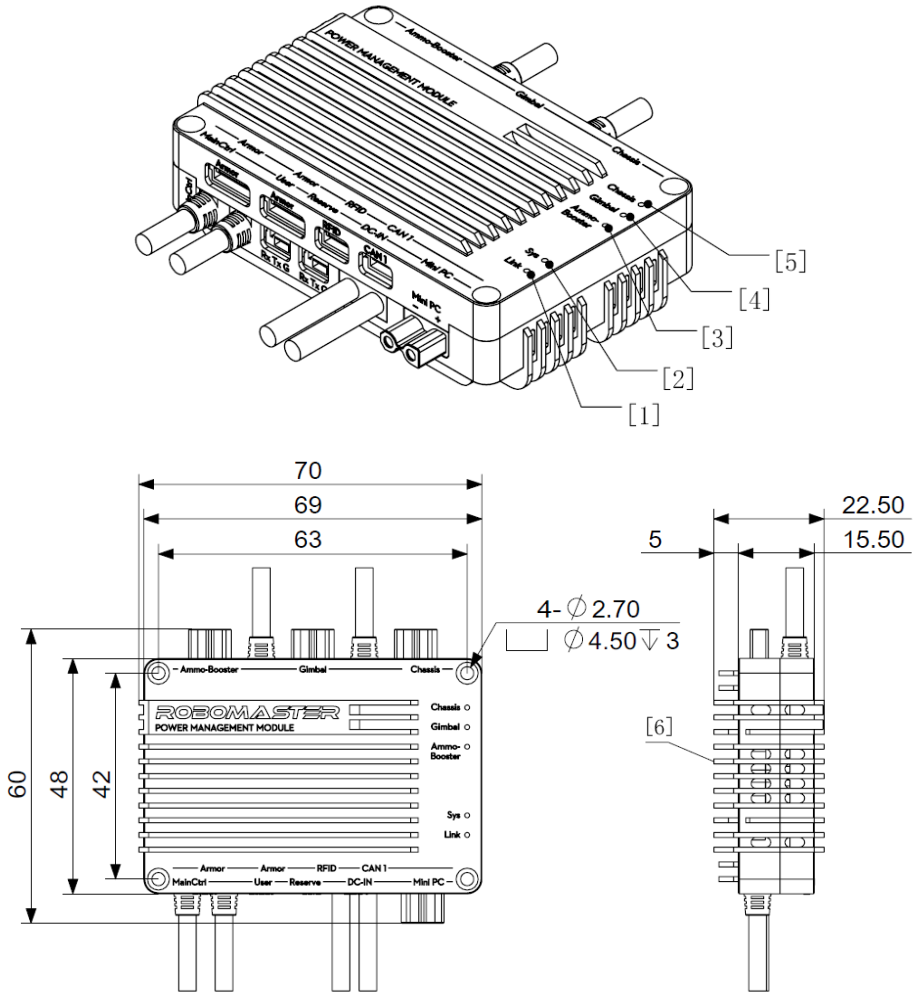


[1] 0.8 m Referee System cable with female connector [2] Dart System placement surface

Figure 3-6 Dart Launching Station Referee System aviation cable position diagram

3.4 Power Management Module Installation Specifications

Refer to the Power Management Module dimensions and reserve mounting holes at a specific position on the robot.



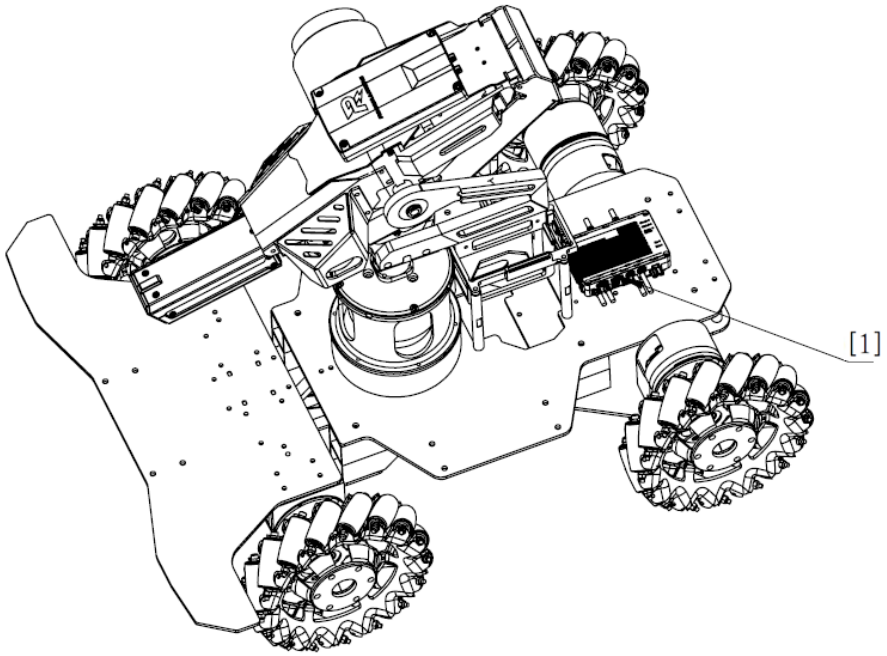
[1] Connection status indicator	[2] System status indicator	[3] Launching mechanism power output indicator
[4] Gimbal power output indicator	[5] Chassis power output indicator	[6] Bottom mounting surface

Figure 3-7 Power Management Module diagram

3.4.1 Installation Steps

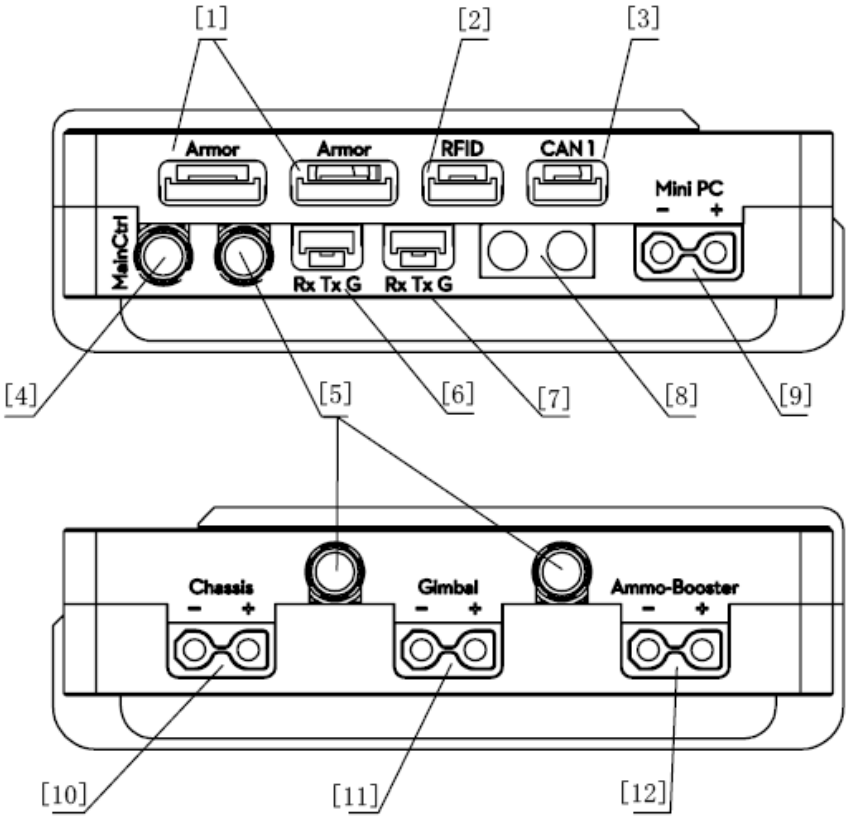
The aviation cable interfaces on the Light Indicator Module, Camera Video Transmitter Module (Transmitter), Speed Monitor Module, and Positioning Module are equivalent interfaces and may be connected to one another in series.

Use four M2.5 screws to secure the Power Management Module to the robot.



[1] Power Management Module

Figure 3-8 Power Management Module installation diagram



[1] Armor Module SM06B-GHS-TB interface

[2] Field Interaction Module SM04B-GHS-TB interface

[3] Supercapacitor Management Module SM04B-GHS-TB interface

[4] Main Controller Module interface (the metal ring on this aviation cable interface is black)

[5] Interfaces for other Referee System onboard-end modules (Speed Monitor, Positioning, video transmitter, and light indicator; the metal ring on these aviation cable interfaces is silver-white)

[6] Referee System serial port SM03B-GHS-TB interface

[7] System upgrade SM03B-GHS-TB interface

[8] Referee System power XT60 interface (input)

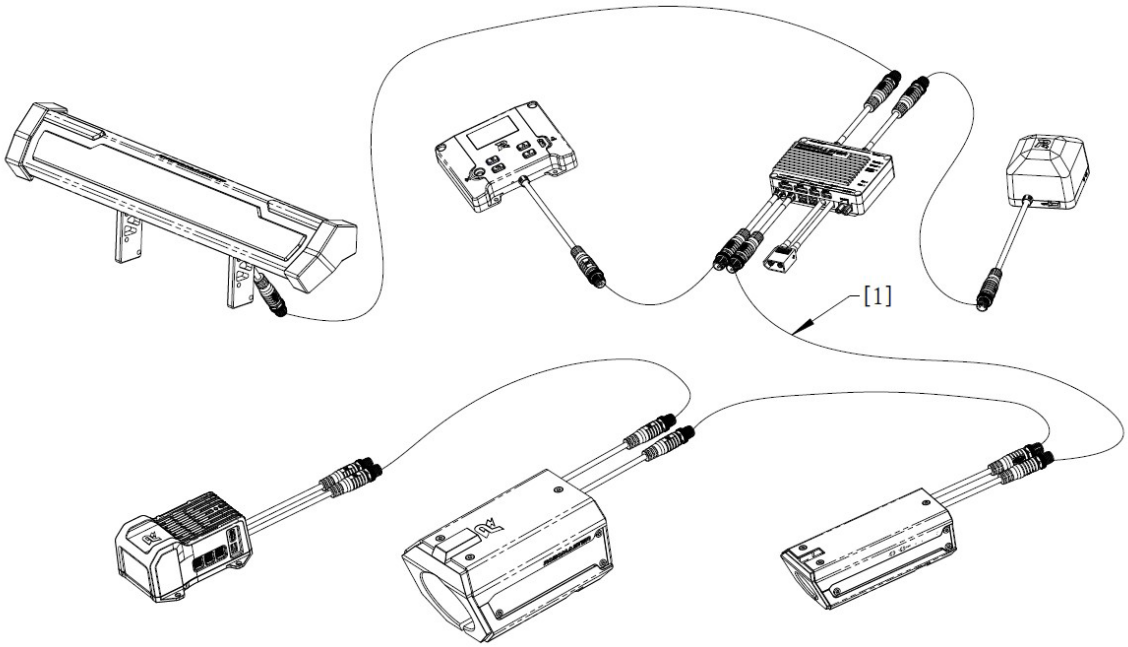
[9] Mini PC power XT30 interface (output)

[10] Referee System power XT30 interface (output) — connects to chassis

[11] Referee System power XT30 interface (output) — connects to Gimbal

[12] Referee System power XT30 interface (output) — connects to launching mechanism

Figure 3-9 Power Management Module interface diagram



[1] Aviation adapter cable

Figure 3-10 Power Management Module wiring diagram

3.4.2 Installation Requirements

The Power Management Module installation must meet these requirements:

Securely fasten the Power Management Module to heat-resistant material in a well-ventilated location, and protect all connectors. The module heats up during use. If secured with adhesive such as double-sided tape, it may detach and create a safety hazard. The team bears responsibility for consequences caused by improper fastening or inadequate heat dissipation.

S111 For robots subject to the Chassis Power limit, the power used by mechanisms that count toward Chassis Power must not bypass monitoring by the Power Management Module.

S112 The Dart System and Radar System must use the Main Controller Module installed at the field by RMOC. Therefore, teams must meet these requirements when designing their robots:

- When designing the Dart System or Radar System structure, expose the Main Controller Module interface on the Power Management Module.
- When designing the Dart System structure, install the Power Management Module close to the Dart Launching Station entrance.

S113 Carefully identify the Power Management Module interfaces and wire them correctly. Refer to “Table 3-4 Power Management Module Interface Description” for the power connections for each mechanism.

Except for Ground Robot chassis power, other robot power interfaces (such as Gimbal power and 42mm launching mechanism power) may be battery-powered for stable supply and controlled by relays or other means.

but they must be switched on and off through the corresponding Power Management Module interfaces listed in the table below (relays and other means must receive power through the corresponding interface, ensuring that the robot Referee System can control the power to all devices connected to “Referee System power interface (output)”; otherwise, this is considered cheating).

- A device connected to a Power Management Module power interface must shut off immediately when that interface is powered off.
- If the Radar System or Dart Launcher requires a 24 V power interface, it may be powered directly from the Power Management Module “Mini PC” interface or a battery.

Table 3-4 Power Management Module interface description

Interface / Robot type	Chassis	Gimbal	Ammo-Booster	Mini PC
Heavy Robot	Chassis power	Gimbal power	42mm launching mechanism power	Mini PC power
Infantry Robot	Chassis power	Gimbal power	17mm launching mechanism power	Mini PC power
Hangar	No power	Actuator power	No power	Mini PC power
Sentry Robot	Chassis power	Gimbal power	17mm launching mechanism power	Mini PC power
Dart Launcher	No power	No power	No power	Mini PC power
Radar computing platform end	No power	No power	No power	Mini PC power

S114 When the Ammo-Booster interface on the Power Management Module is powered off, the launching mechanism (friction-wheel motors, projectile-feeding disk motors, solenoid valves, etc.) on a robot with a launching mechanism must also be powered off.

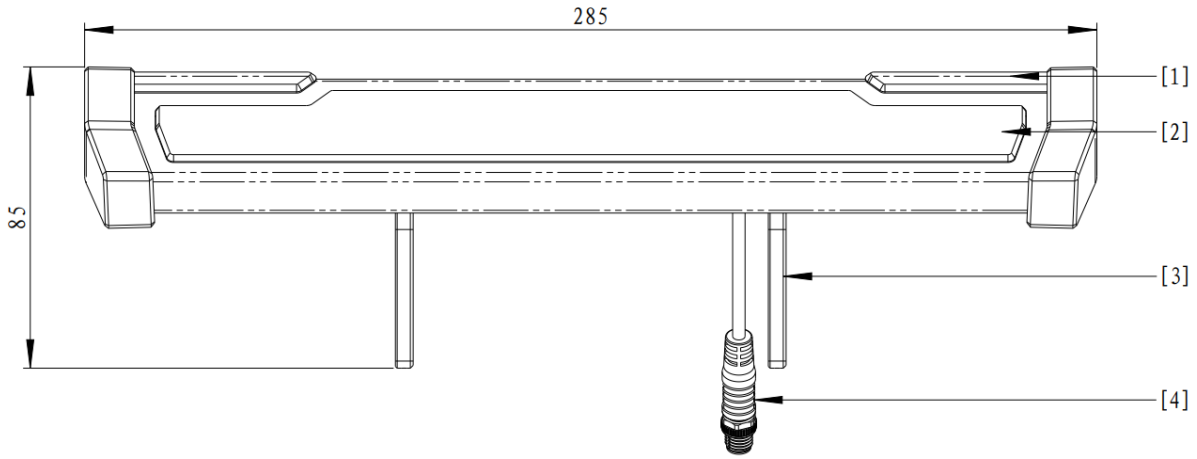
S115 For robots subject to the Chassis Power limit, circuit boards and wiring must meet the following requirements:

- Circuit boards related to chassis power must be independent of Gimbal and launching-mechanism power. A circuit board powered through the Power Management Module “Chassis” interface must not connect to any other Power Management Module power interface.
- All robot chassis wiring must be clear. Referees will spot-check robots after matches, and teams must cooperate by disassembling relevant robot parts and showing the wiring. Teams are advised to account for these inspections when designing and routing wiring; the team bears any lost preparation time caused by disassembling the robot for inspection.
- A circuit board connected to the Power Management Module “Chassis” interface (a chassis-power-related board) may connect to a board powered from another Power Management Module power port only through cables no larger than 24 AWG. Such connections are for communication only, and total current must not exceed 50 mA.

- The Power Management Module input voltage must be 22–26 V. Power output interfaces 10, 11, and 12 in “Figure 3-9 Power Management Module Interface Diagram” can be switched on and off by the Referee System.

- For the load thresholds of each Power Management Module interface, see the Power Management Module Manual.
- When system load fluctuates significantly, the voltage of power output interfaces 9–12 will fluctuate. Teams are advised to stabilize voltage-sensitive loads (such as a Mini PC).

3.5 Light Indicator Module Installation Specifications



[1] Auxiliary light strip [2] Main light strip [3] Mounting bracket [4] Aviation cable interface

Figure 3-11 Light Indicator module diagram

3.5.1 Installation Steps

1. Optional: The Light Indicator Module may be attached using the bottom screw holes of its mounting bracket and installed in a suitable position on the robot.

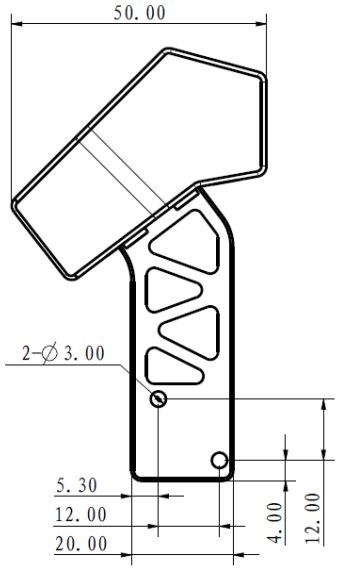


Figure 3-12 Light Indicator module screw installation positions

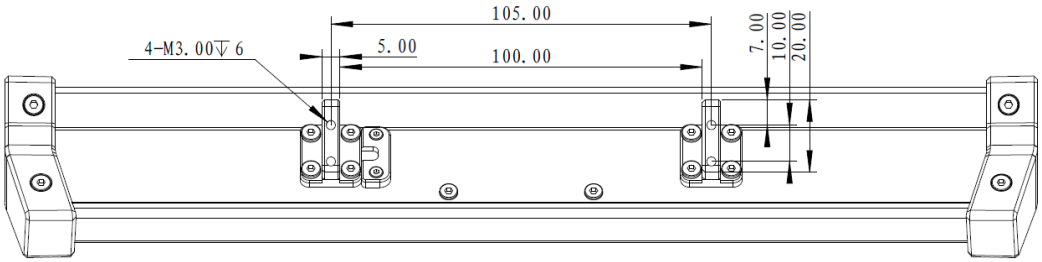
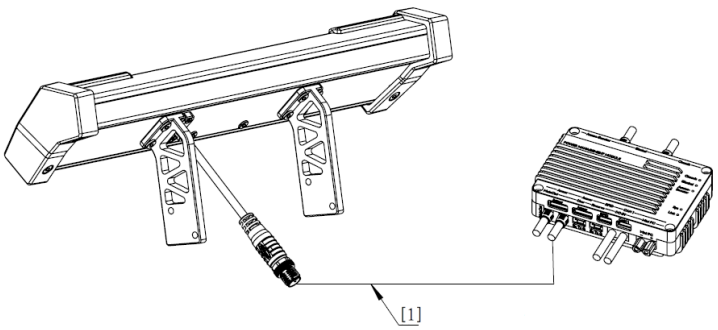


Figure 3-13 Light Indicator module bottom view

2. Use the aviation adapter cable included in the package to connect the Light Indicator Module to the aviation cable interface with the silver-white metal ring on the Power Management Module.



[1] Aviation adapter cable

Figure 3-14 Light Indicator module wiring diagram

3.5.2 Installation Requirements

The Light Indicator Module must be installed in accordance with the following requirements:

S116 The wiring for the left and right auxiliary light strips must be parallel to the ground.



Installation recommendation: From at least one horizontal direction, the main and auxiliary light strips should be fully visible when viewed directly from above or below at a 45° angle. Otherwise, information conveyed by the light effects may be lost during a match, delaying detection and handling of abnormal conditions.

3.6 Armor Module Installation Specifications

Armor Modules must be installed on the robot using the designated Armor Support Brackets.

S117 An Armor Support Bracket may be installed on the robot chassis using only its rear or bottom mounting surface.

S118 Only the official Armor Support Brackets may be used to install Armor Modules. Modifying or damaging the brackets is prohibited.

The designated Armor Support Bracket is shown below:

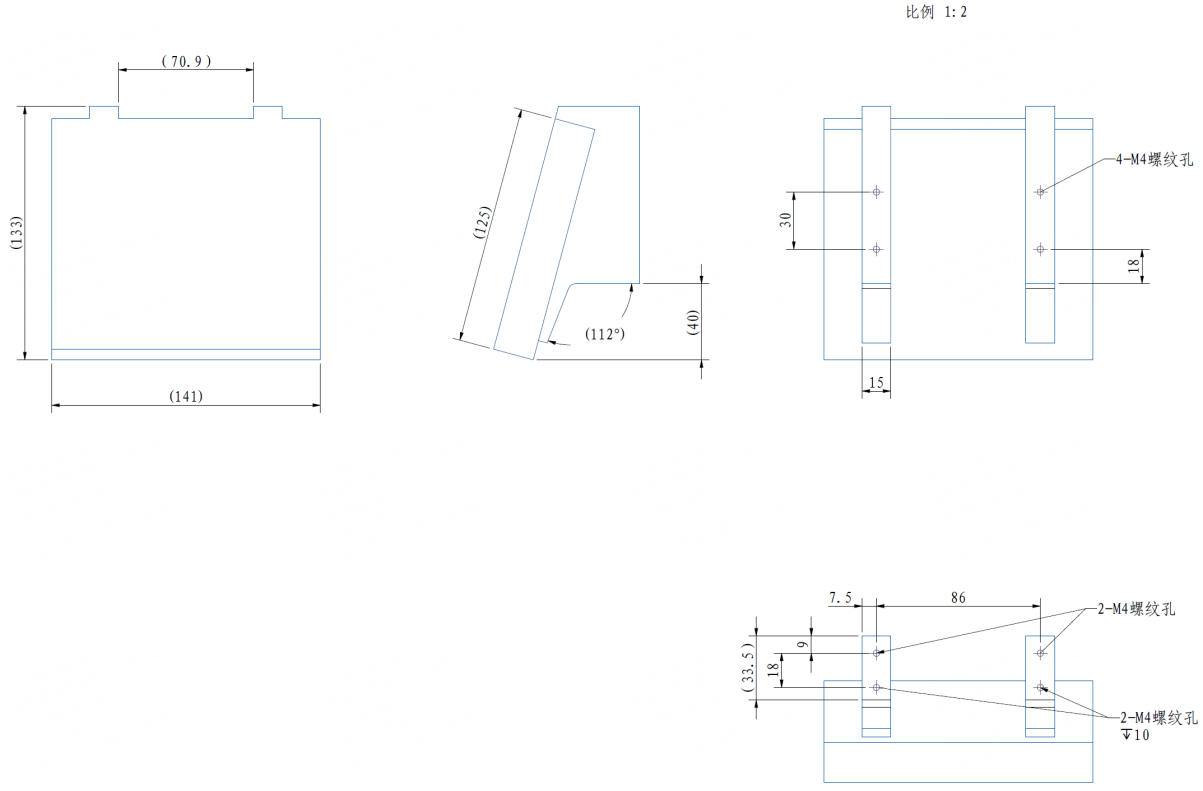


Figure 3-15 Designated armor support bracket diagram

S119 Do not modify or decorate Armor Modules in any way.

3.6.1 General Specifications

Robot chassis: The mechanism that carries the robot's drive system and associated components and supports the robot body.

The robot body coordinate system is a standard Cartesian X, Y, Z coordinate system whose origin is the robot's center of mass. Based on the installation requirements for Ground Robot Armor Modules, define the robot coordinate system with the X axis in the direction of theoretically highest efficiency for the chassis configuration and the positive Z axis pointing toward the center of the Earth. If a special chassis configuration (for example, steering wheels) has multiple maximum-efficiency directions, any one may be chosen as the X axis. The X-axis definitions for different chassis configurations are shown below:

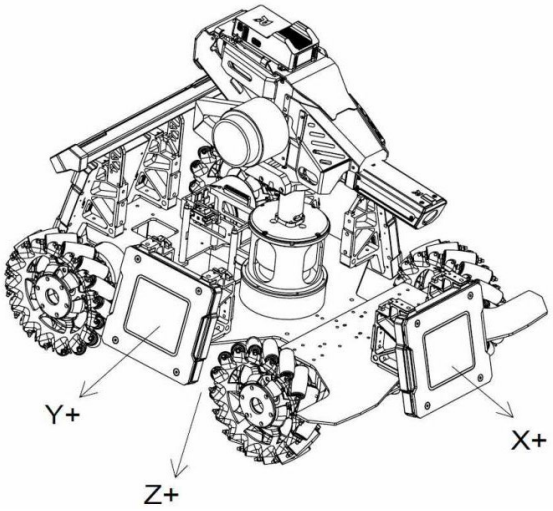


Figure 3-16 Robot coordinate system diagram

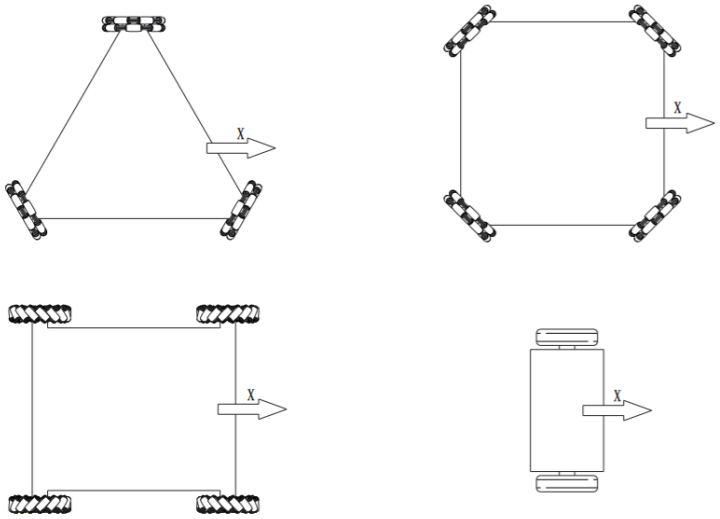


Figure 3-17 X axis diagram for different robot chassis configurations

3.6.1.1 Armor Module Installation

S120 When installing Armor Modules, securely connect each module to its Armor Support Bracket. The bracket's bottom mounting surface must be parallel to the XY plane so that the acute angle between the line normal to the Armor Module hit-surface plane and the line along the negative Z axis is 75°. The angle between the two Armor Module edges without side lights and the XY plane must be less than 5°. The projection onto the XY plane of the installed Armor Module hit-surface normal (which forms an acute angle with the negative Z axis) is defined as its direction vector. The direction vectors of the four Armor Modules must correspond one-to-one with the positive X, negative X, positive Y, and negative Y axes of the robot body coordinate system; the angular error between each vector and its corresponding axis must not exceed 5°.

S121 The robot's own kinematic equations must also be based on the body coordinate system above. Armor Module installation and the robot's structural or kinematic characteristics must share the same reference coordinate system. The line connecting the geometric centers of the Armor Modules on the X axis must be perpendicular to the line connecting the geometric centers of those on the Y axis. After displacement, the X-axis component of the offset between the geometric center of an Armor Module on the Y axis and the robot's geometric center must not exceed 50 mm; the same applies to an Armor Module on the X axis.

3.6.1.2 Rigid Connection

S122 The installed Armor Modules and Armor Support Brackets must be rigidly connected to the chassis. An Armor Module must not move relative to the chassis during a match. As shown below, rigid connection is defined as follows: when a 60 N vertically upward force is applied to the midpoint of an Armor Module's lower edge, the angle α of its hit surface must change by no more than 2.5°.

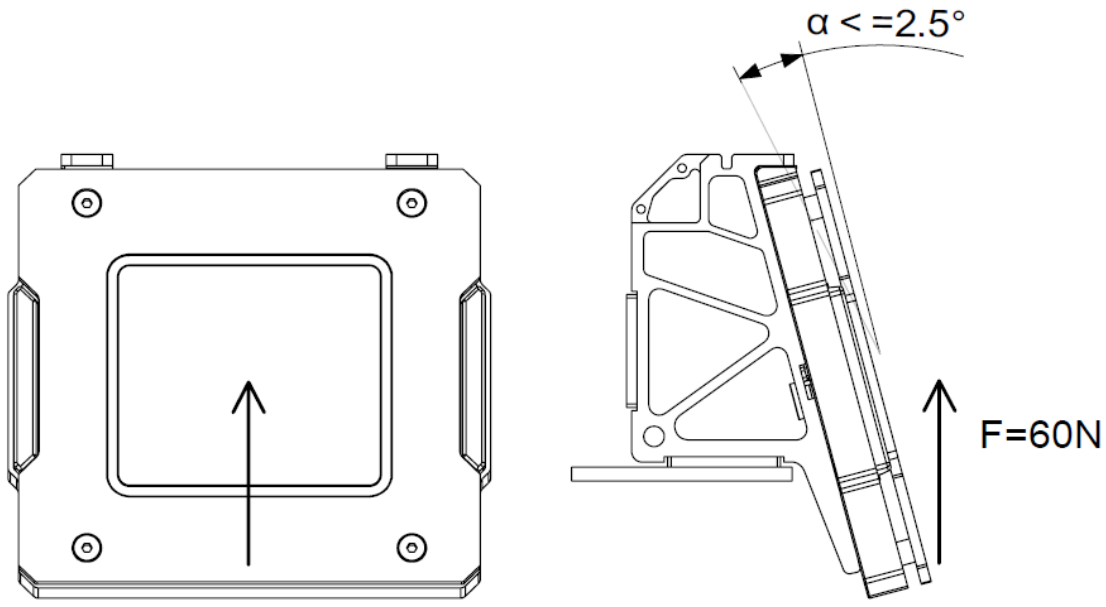


Figure 3-18 Armor Module force diagram

3.6.1.3 Robot Deformation

S123 In principle, after a match begins, no Armor Module may actively move relative to the robot's overall center of mass. If

A robot may be deformable because of its structural design. In that case, its Armor Modules must never move continuously, reciprocally, and rapidly relative to the chassis. Rapid movement is defined as a speed exceeding 0.5 m/s.

S124 The height difference between the lower edges of any two Armor Modules on a Ground Robot must not exceed 100 mm. Temporary noncompliance is permitted while the robot crosses terrain.

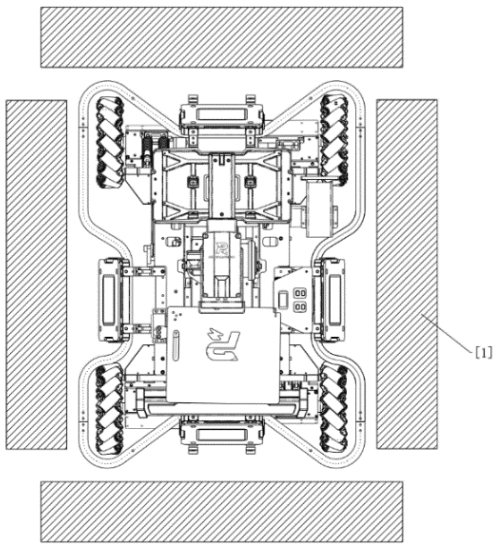
S125 For Ground Robots, before, during, and after transformation, the lower edge of an Armor Module must be no more than 400 mm above the ground and must not directly contact the ground.

- If the surface supporting the robot body is not the ground, measure the Armor Module installation height from the highest plane supporting the robot body.
- A robot with no supporting plane is not subject to this requirement.

3.6.1.4 Armor Module Protection

S126 Teams must design bumpers for Ground Robots to reduce damage caused by collisions with Armor Modules.

S127 When any side of the robot is placed close against a vertical rigid plane on level ground, no Armor Module may directly contact that plane, as shown below:



[1] Wall

Figure 3-19 Robot protection diagram

S128 No structure other than the Armor Support Bracket may contact an Armor Module.

3.6.2 Installation Steps

Ground Robot Armor Modules use the same installation steps. The following steps use the Infantry Robot Armor Module as an example. The steps below

use only the bottom mounting surface of the Armor Support Bracket.

1. Reserve four sets of mounting holes on the chassis, one set for each Armor Module. Keep the size and relative position of the four holes in each set consistent.

2. Secure each Armor Support Bracket to the chassis with two M4 screws. The completed installation is shown below:

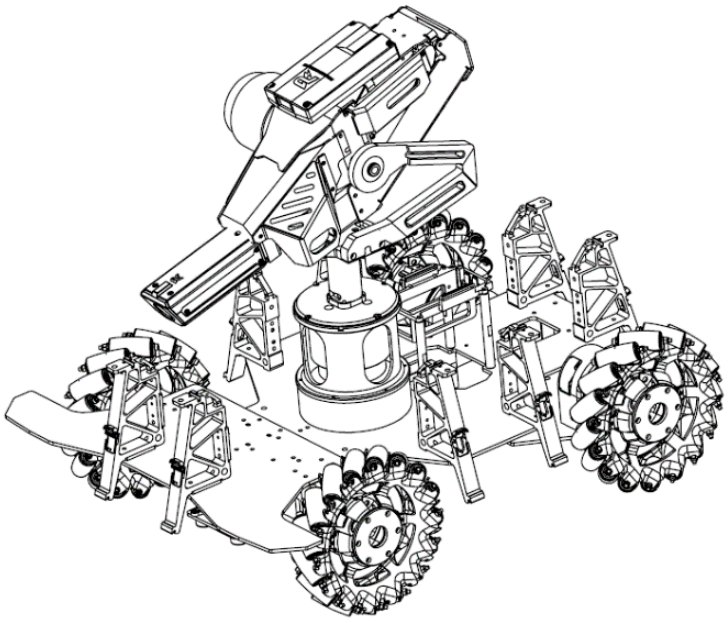


Figure 3-20 Armor support bracket installation diagram

3. Install the Armor Module on the Armor Support Bracket and secure it with M4 screws. Installation steps follow:

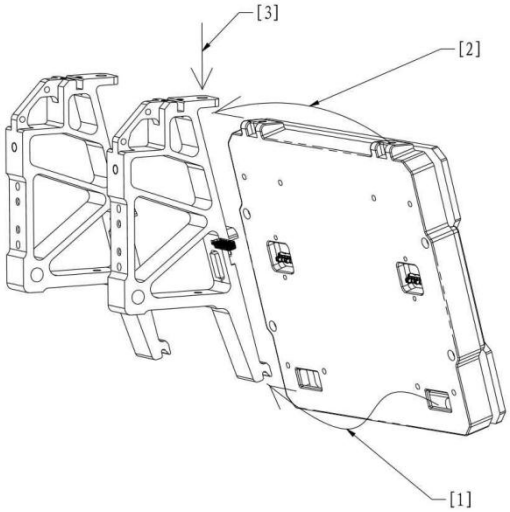


Figure 3-21 Armor Module installation diagram

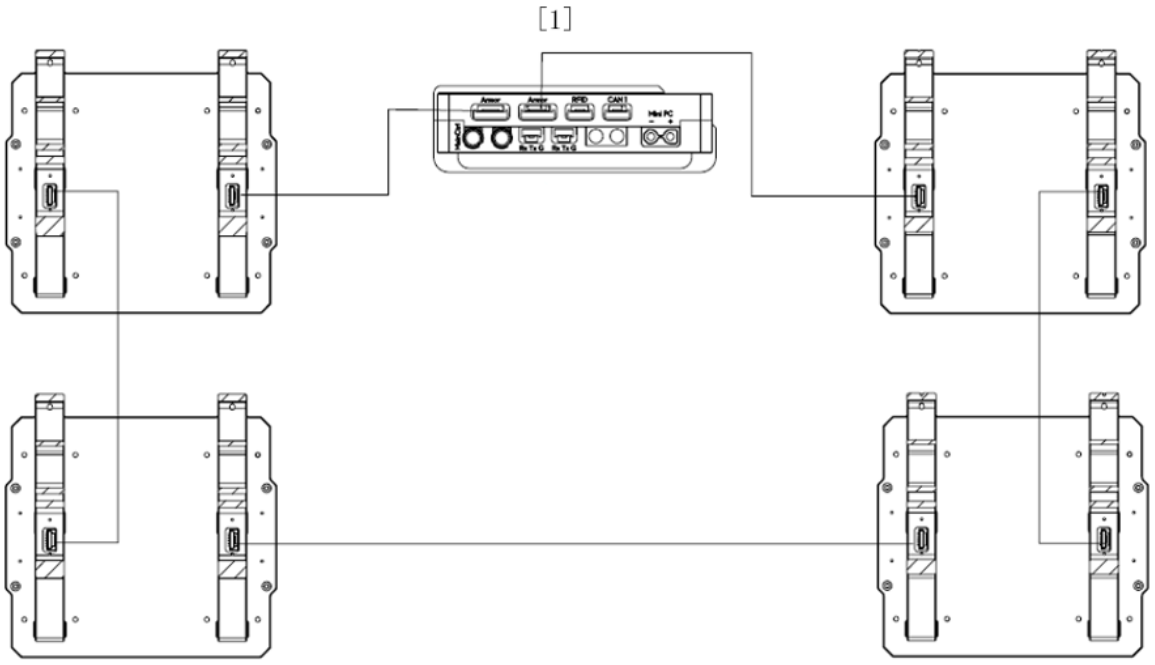
1) Insert the lower slot of the Armor Module into the lower hook on the Armor Support Bracket.

2) Insert the upper end face of the Armor Module into the upper hook on the Armor Support Bracket.

3) Insert the fastening screw.

4. Use the 6-pin cables provided in the package to connect each Armor Module in series to the Armor Module interface on the Power Management Module. The two 6-pin interfaces on the Armor Support Bracket are equivalent.

- Route the cables appropriately for the robot, ensuring secure connections and preventing cable damage.
- When connecting the modules, it is recommended to distribute them evenly between the two 6-pin interfaces on the Power Management Module so the current is shared evenly.



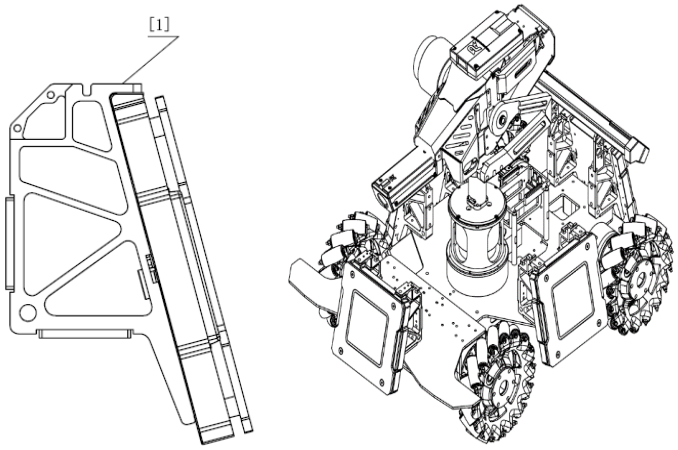
[1] Power Management Module

Figure 3-22 Armor Module wiring diagram

The specific installation steps will be updated later.

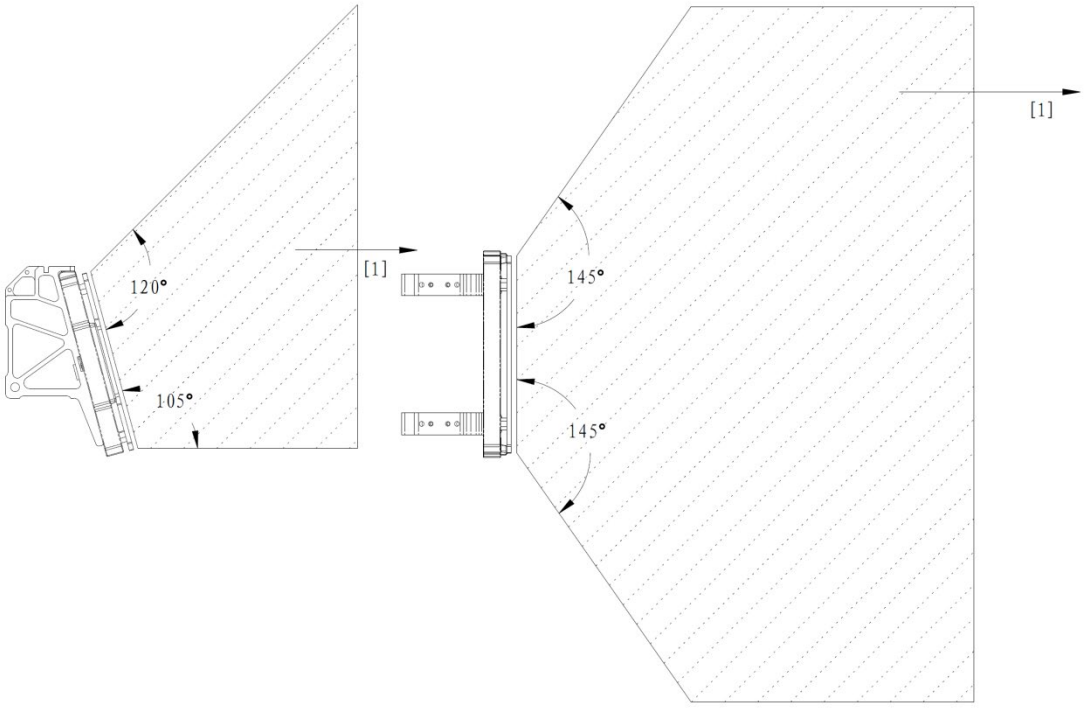
3.6.3 Installation Requirements

S129 The hit surfaces of Infantry Robot and Sentry Robot Armor Modules must be unobstructed within 105° of the lower edge, 120° of the upper edge, and 145° of the left and right edges.



[1] Secure at the top with an M4 screw

Figure 3-23 Armor Module installation specification diagram



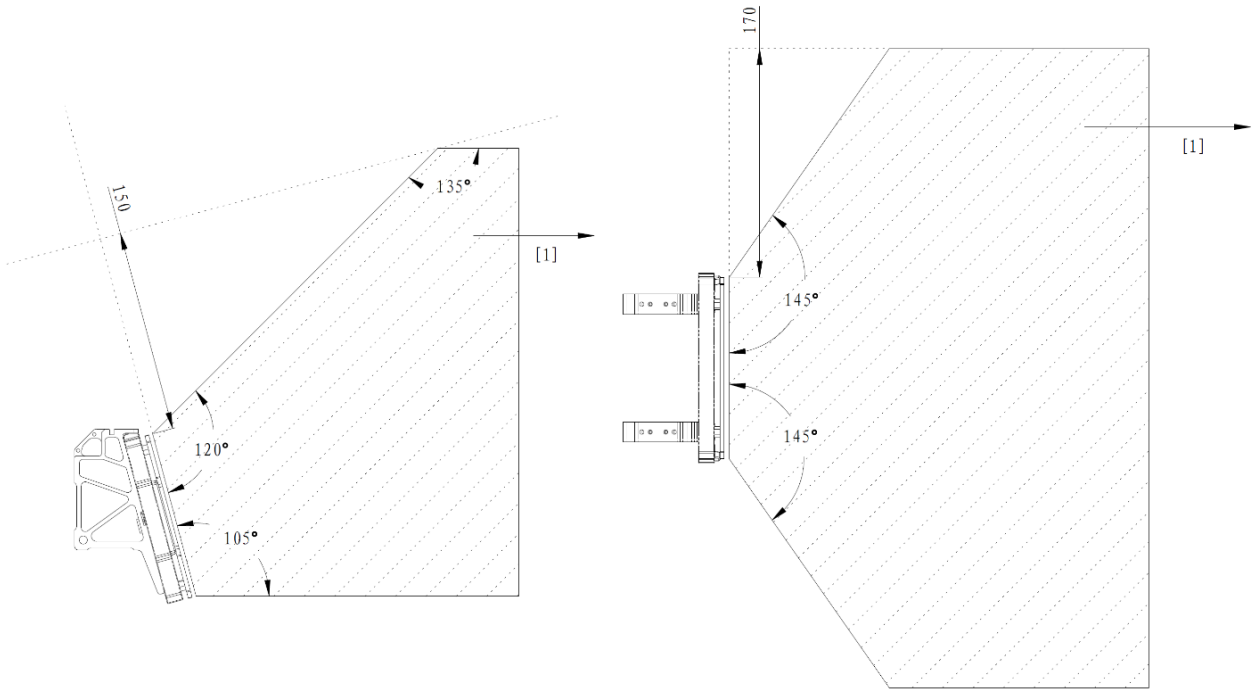
[1] Extends infinitely

Figure 3-24 No-obstruction zone of Infantry Robot and Sentry Robot Armor Modules

S130 The hit surfaces of all Heavy Robot Armor Modules must be unobstructed within 105° of their lower edges. At least three of the four Armor Modules

must also be unobstructed within 120° of the upper edge and 145° of the left and right edges. At most one Armor Module may be conditionally obstructed within 120° of the upper edge or 145° of the left and right edges of its hit surface. Obstruction is permitted at points more than 150 mm from the upper edge or more than 170 mm from the left or right edge, measured in the plane of the hit surface; the shaded region in the figure below must remain unobstructed.

The prohibited-obstruction zone in this section extends infinitely within the robot's maximum extended dimensions.



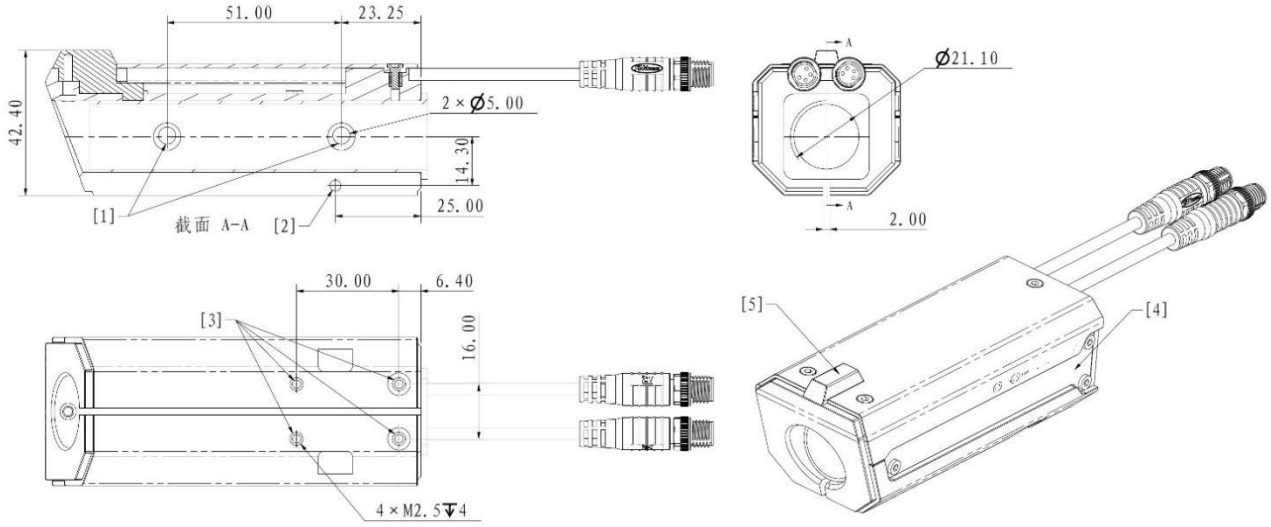
[1] Extends infinitely

Figure 3-25 No-obstruction zone of Heavy Robot Armor Modules

3.7 Speed Monitor Module Installation Specifications

There are two types of Speed Monitor Module: the 17mm Speed Monitor Module and the 42mm Speed Monitor Module.

17mm Speed Monitor Module:



[1] Phototube

[2] Speed Monitor Module clamping screw hole

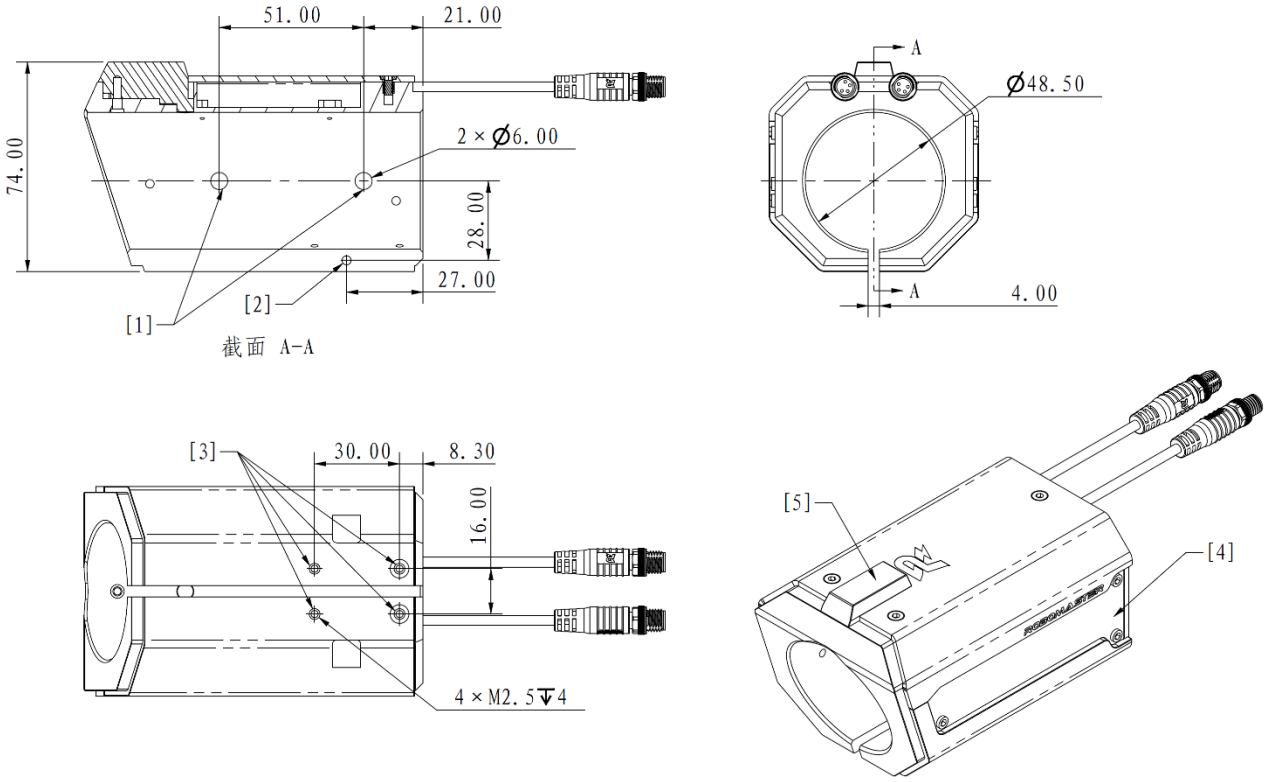
[3] M2.5 mounting screw hole

[4] LED light strip

[5] Sight

Figure 3-26 17mm Speed Monitor Module diagram

42mm Speed Monitor Module:



[1] Phototube

[2] Speed Monitor Module clamping screw hole

[3] M2.5 mounting screw hole

[4] LED light strip

[5] Sight

Figure 3-27 42mm Speed Monitor Module diagram

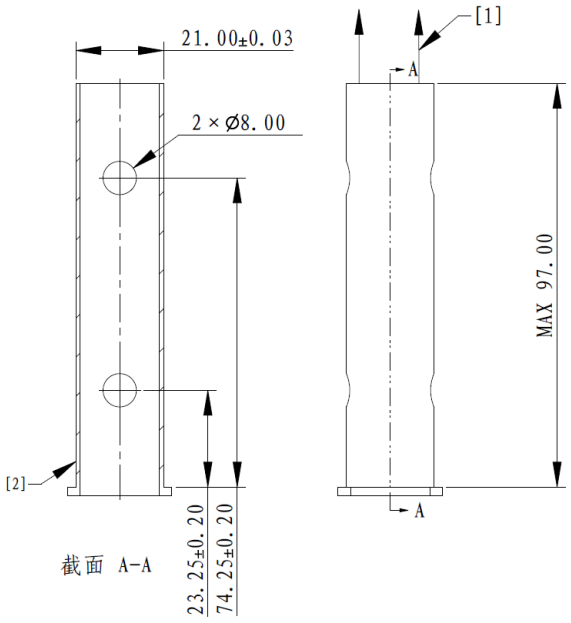
S131 Do not remove the sight from the Speed Monitor Module.

3.7.1 Installation Steps

There are currently three mounting methods for the 17mm Speed Monitor Module. All three comply with the Referee System installation specifications. Teams may choose any one method.

3.7.1.1 Mounting Method 1 for the 17mm Speed Monitor Module

17mm barrel dimensional limits:



[1] Toward the muzzle [2] Recommended wall thickness: at least 1 mm

Figure 3-28 17mm launch tube diagram

17mm barrel construction requirements:

S132 Ensure the phototube is not obstructed.

S133 Transparent or light-emitting materials are prohibited, as are infrared sensors near the barrel.

S134 Ensure the inner wall of the barrel has no highly reflective material, which could interfere with normal operation of the Speed Monitor Module.

Recommendation: Give the inner wall of the barrel a frosted or matte finish.

Installation steps for Mounting Method 1:

1. Place the Speed Monitor Module over the barrel.
2. Pass an M3 screw through the Speed Monitor Module clamping screw hole and tighten it around the barrel.

3. Connect the aviation cable interface on the Speed Monitor Module to the aviation cable interface on the Power Management Module using an aviation adapter cable.

After installation, prevent the barrel from rotating or shifting out of position and obstructing the phototube.

The completed installation is shown below:

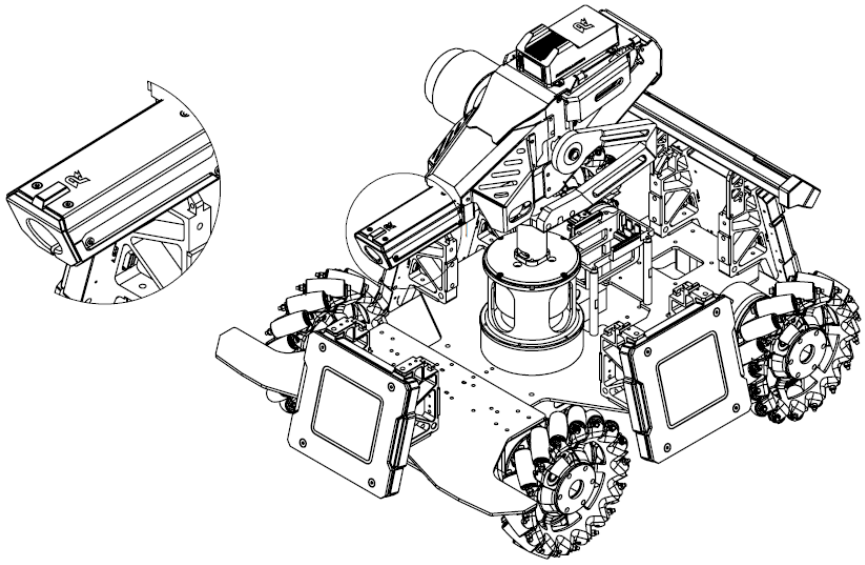


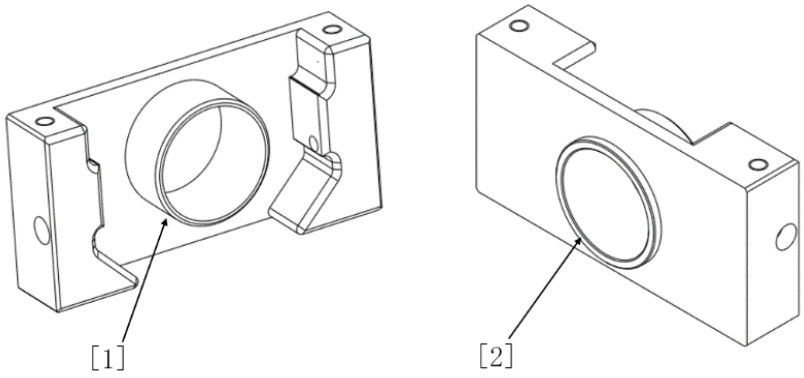
Figure 3-29 Speed Monitor Module installation diagram

3.7.1.2 Mounting Method 2 for the 17mm Speed Monitor Module

Teams must design and fabricate an adapter part to connect the 17mm Speed Monitor Module to the launching mechanism.

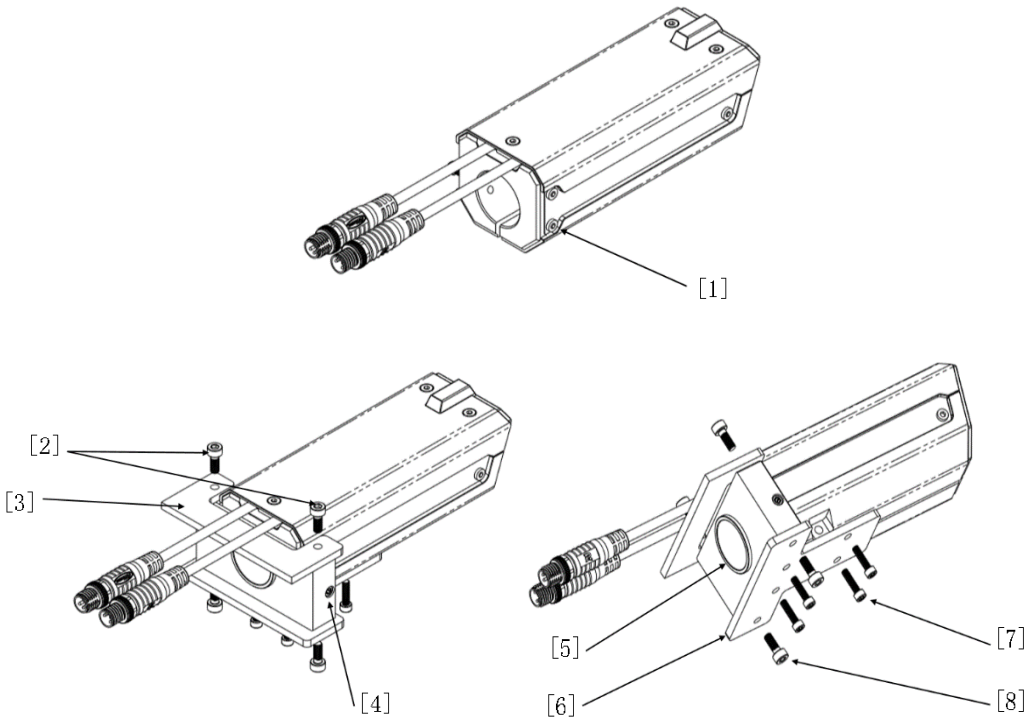
For the adapter part engineering drawing, see “Appendix 1 17mm Speed Monitor Module Adapter Part Engineering Drawing.” A 3D model is available on the Speed Monitor Module product page of the RoboMaster website for reference only.

A reference adapter part is shown below:



[1] Front boss [2] Rear boss

Figure 3-30 17mm adapter part diagram



[1] M2.5 LED light-strip mounting screw	[2] M3 screw	[3] Custom part 1	[4] Side mounting hole
[5] Rear boss	[6] Custom part 2	[7] M2.5 screw	[8] M3 screw

Figure 3-31 17mm adapter part fixing method diagram

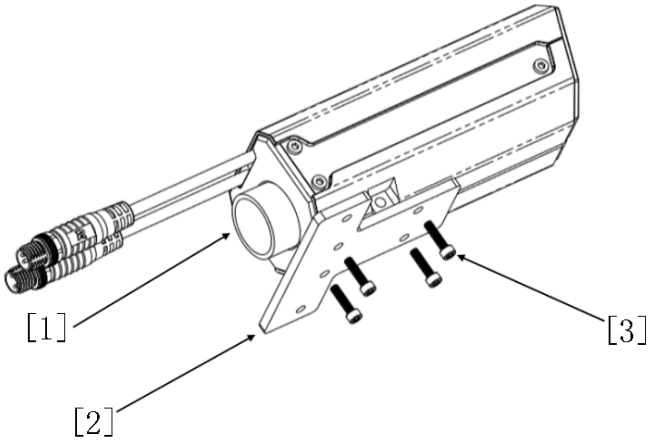
Installation steps for Mounting Method 2:

1. Remove one M2.5 screw securing the LED light strip on each side of the Speed Monitor Module; the position on one side is shown as [1]. Do not remove any other LED light-strip mounting screws on the module.
2. Use two M2.5 × 14 screws through the side mounting holes on both sides (one side is shown as [4]) to secure the adapter part to the Speed Monitor Module.
3. Use two M3 screws to secure the robot's existing plate part 1 to the top of the Speed Monitor Module.
4. Use two M3 screws and four M2.5 screws to secure the robot's existing plate part 2 to the bottom of the Speed Monitor Module.
5. Connect the aviation cable interface on the Speed Monitor Module to the aviation cable interface on the Power Management Module using an aviation adapter cable.

Except for the two screws permitted to be removed in Step 1, do not remove or install any other screws on the Speed Monitor Module without authorization. Violation is considered damage to the Referee System.

3.7.1.3 Mounting Method 3 for the 17mm Speed Monitor Module

Teams must design and fabricate a barrel part to connect the 17mm Speed Monitor Module to the launching mechanism.



[1] Barrel [2] Custom part 1 [3] M2.5 screw

Figure 3-32 17mm short launch tube installation diagram

Installation steps for Mounting Method 3:

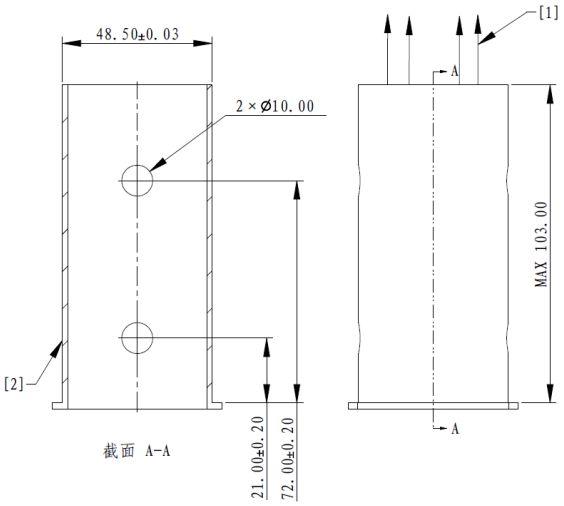
1. Insert the Speed Monitor Module into the short barrel.
2. Use four M2.5 screws to secure the robot's existing plate part 1 to the bottom of the Speed Monitor Module.
3. Connect the aviation cable interface on the Speed Monitor Module to the aviation cable interface on the Power Management Module using an aviation adapter cable.

- The part of the barrel inserted into the Speed Monitor Module must be no longer than 23 mm, to avoid obstructing the module's phototube.
- The recommended outer diameter of the barrel is 21 mm + 0.05. If the barrel diameter is too small, the gap between its outer wall and the module's inner wall will be large, increasing projectile dispersion.
- This method does not locate the Speed Monitor Module relative to the launching-mechanism part, so the module axis may not align with the projectile launch axis, causing some projectiles to hit the module's inner wall. If needed, add shims between the robot's existing plate part 1 and the Speed Monitor Module to adjust their mounting angle.

3.7.1.4 Mounting Method for the 42mm Speed Monitor Module

The three mounting methods for the 17mm Speed Monitor Module may also be used as references for mounting the 42mm Speed Monitor Module.

42mm barrel dimensional limits:



[1] Toward the muzzle

[2] Wall thickness must be at least 1 mm

Figure 3-33 42mm launch tube diagram

42mm barrel construction requirements:

S135 Ensure the phototube is not obstructed.

S136 Transparent or light-emitting materials are prohibited, as are infrared sensors near the barrel.

S137 Ensure the inner wall of the barrel has no highly reflective material, which could interfere with normal operation of the Speed Monitor Module.

Recommendation: Give the inner wall of the barrel a frosted or matte finish.

Mounting steps:

1. Place the Speed Monitor Module over the barrel.
2. Pass an M3 screw through the Speed Monitor Module clamping screw hole and tighten it around the barrel.
3. Connect the aviation cable interface on the Speed Monitor Module to the aviation cable interface on the Power Management Module using an aviation adapter cable.

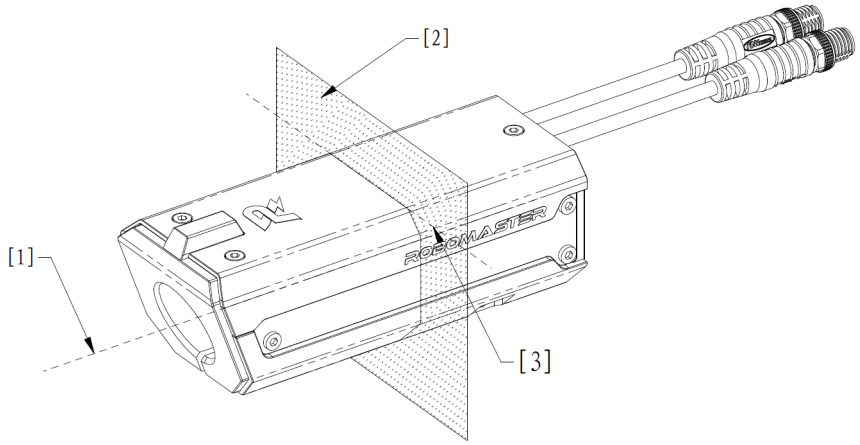
3.7.2 Installation Requirements

The Speed Monitor Module installation must meet the following requirements:

Recommendation: For each Speed Monitor Module, at least one LED-strip emitting-surface normal should point into the upper hemisphere above the horizontal plane. An inspector standing at least 1 m away and directly facing the LED-strip emitting surface should be able to see 80% of the strip. Otherwise, information conveyed by the light effects may be lost during a match, delaying detection and handling of abnormal conditions.

S138 Install the Speed Monitor Module at the end of the launching mechanism to ensure the projectile enters the module only after it has been fully accelerated.

S139 When installing the Speed Monitor Module, the intersection of the plane perpendicular to the projectile launch direction and the Logo plane must be horizontal, as shown below. Maintain this requirement while launching projectiles; an angle of up to 5° relative to the ground is permitted.



[1] Projectile direction [2] [3] Intersection of the Logo plane and [2]

Figure 3-34 Speed Monitor Module installation diagram

S140 When leveling the Speed Monitor Module, its Logo surface must face upward.

S141 Secure the Speed Monitor Module firmly. It must not move relative to the preset projectile immediately adjacent to the launched projectile inside the launching mechanism.

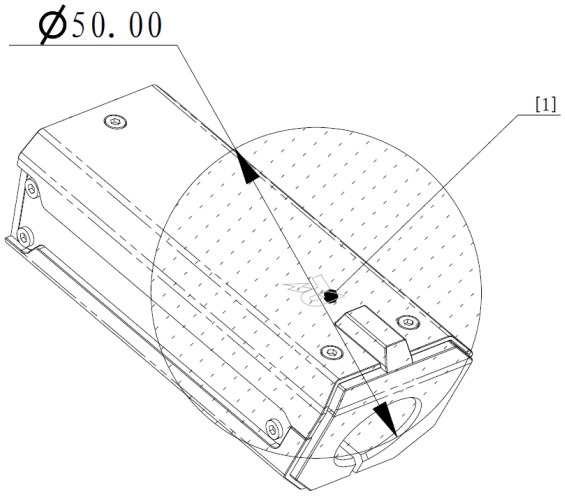
S142 The barrel installed inside the Speed Monitor Module must not be made of iron or other materials that are easily magnetized.

S143 For strongly magnetic materials or equipment that generates a strong magnetic field while operating, the effect on the Speed Monitor Module's magnetic angle while stationary or operating must be within 2° (this requirement does not apply to RMUL).

After calibrating the Speed Monitor Module, briefly press the confirm button on the Main Controller Module, then briefly press the down button. Observe changes in the robot's real-time coordinates on the Main Controller Module as follows (for details, see the onboard-end section of the Referee System User Manual):

- Rotate the Gimbal by 90°, 180°, and 240° respectively; the error compared with the actual rotation angle must be less than 15°.
- For strongly magnetic materials installed in accordance with the requirements above, observe whether the Speed Monitor Module angle fluctuates within 2°.
- For equipment that generates a strong magnetic field while operating, power it on and observe it in both stationary and operating states (for example:
  the friction-wheel motor powered on while stationary and while rotating), and check whether the Speed Monitor Module angle fluctuates within 2°.

S144 As shown in the Speed Monitor Module installation diagram, with the Logo position as the center, no easily magnetized materials (iron, cobalt, nickel products, or components such as the cooling fan of a Camera Video Transmitter Module (Transmitter) or an industrial camera body) may be used within a 50 mm diameter. Within a 100 mm diameter, no magnetic materials other than the Camera Video Transmitter Module or industrial camera body (such as friction-wheel motors or Gimbal motors) may be used, to avoid interfering with the magnetometer inside the Speed Monitor Module.



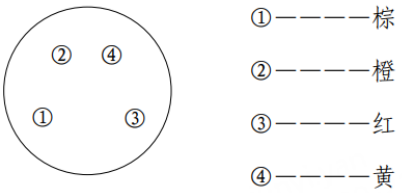
[1] Center position

Figure 3-35 Speed Monitor Module installation specification diagram

- Four M2.5 threaded holes are provided.
- Do not obstruct the phototube opening; doing so will cause the Speed Monitor Module self-check to fail.
- The Speed Monitor Module aviation cable is close to the friction wheels; protect it from wear during use.

S145 Teams must ensure electrical isolation between the modular 42mm launching mechanism and the Heavy Robot as follows:

- For the four-pin Referee System aviation-cable connector from the Heavy Robot, short any two pins together without an external power supply connected. The Heavy Robot Referee System must operate normally.
- For the four-pin Referee System aviation-cable connector from the modular 42mm launching mechanism, with an external power supply of no more than 30 V connected, connect the external supply's positive and negative terminals to any two pins. The launching mechanism Referee System may go offline but must not be damaged.



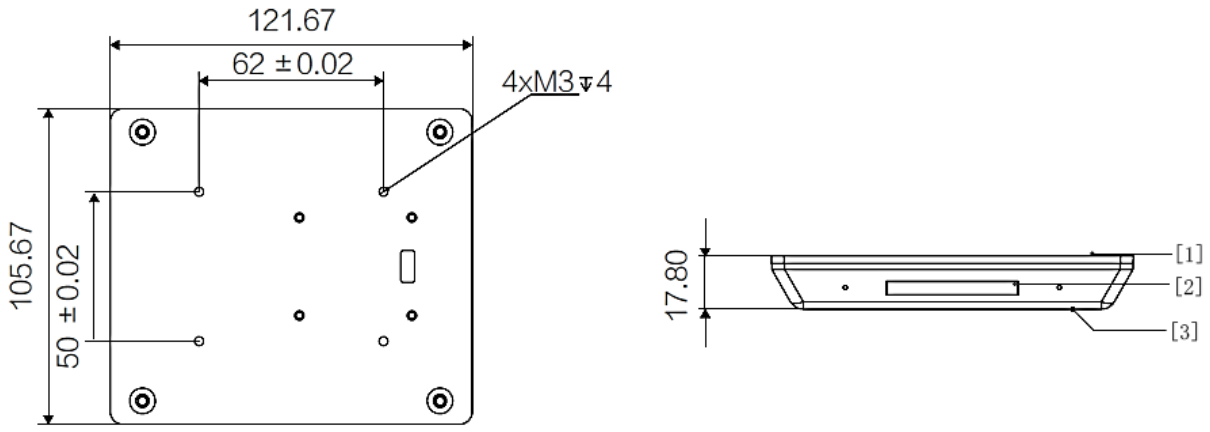
[1] Brown—GND [2] Orange—CAN_L [3] Red—12 V power [4] Yellow—CAN_H

Figure 3-36 Referee System wire sequence diagram

The team must perform these procedures itself during the actual Pre-Match Inspection.

3.8 Field Interaction Module Installation Specifications

Refer to the Field Interaction Module dimensions and mounting interfaces, and reserve mounting holes on the robot chassis.



[1] Back [2] LED light strip [3] Front

Figure 3-37 Field Interaction Module diagram

3.8.1 Installation Steps

1. Use the 4-pin cable provided in the package to connect the Field Interaction Module to the RFID interface on the Power Management Module.

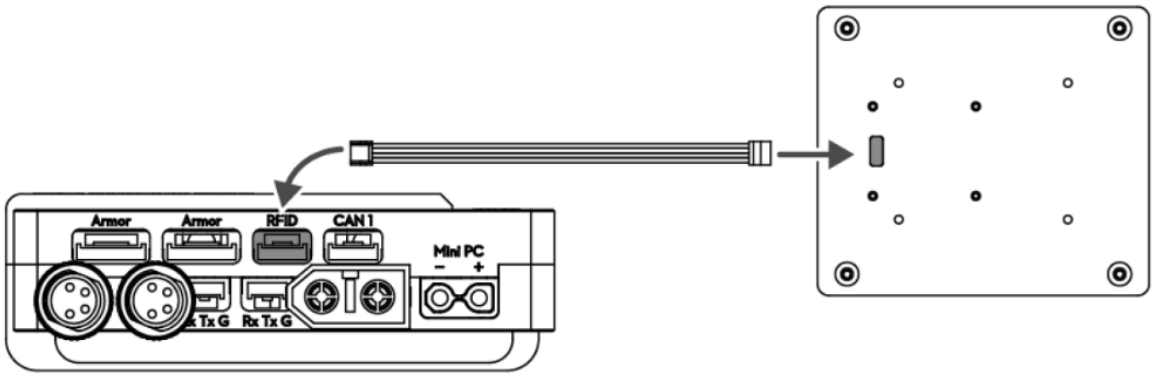


Figure 3-38 Field Interaction Module wiring diagram

2. Use M3 screws to secure the Field Interaction Module to the chassis. Do not pinch the cable during installation, and maintain appropriate clearance between the module and the ground.

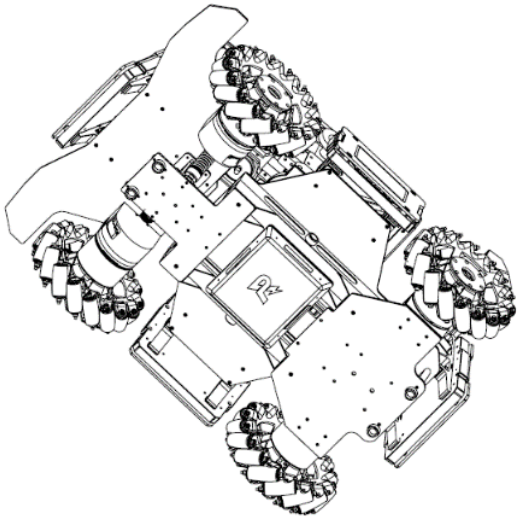


Figure 3-39 Field Interaction Module installation diagram

3.8.2 Installation Requirements

Install the Field Interaction Module in accordance with these requirements; otherwise, its detection quality may decline. The team bears responsibility for consequences caused by improper installation.

- Avoid installing large areas of conductive material near the Field Interaction Module, and avoid high-current or high-frequency signal interference (such as motor wires, RoboMaster central boards, CAN wires, or Supercapacitors).
- Do not obstruct the front or back of the Field Interaction Module with conductive material. Keep at least 30 mm between its back and conductive materials such as metal or carbon-fiber plates.

S146 The Field Interaction Module must not extend beyond the outer boundary of the rectangle formed by the Armor Modules. During robot transformation, it may move relative to the chassis only along the Z axis, with no relative displacement in any other direction.

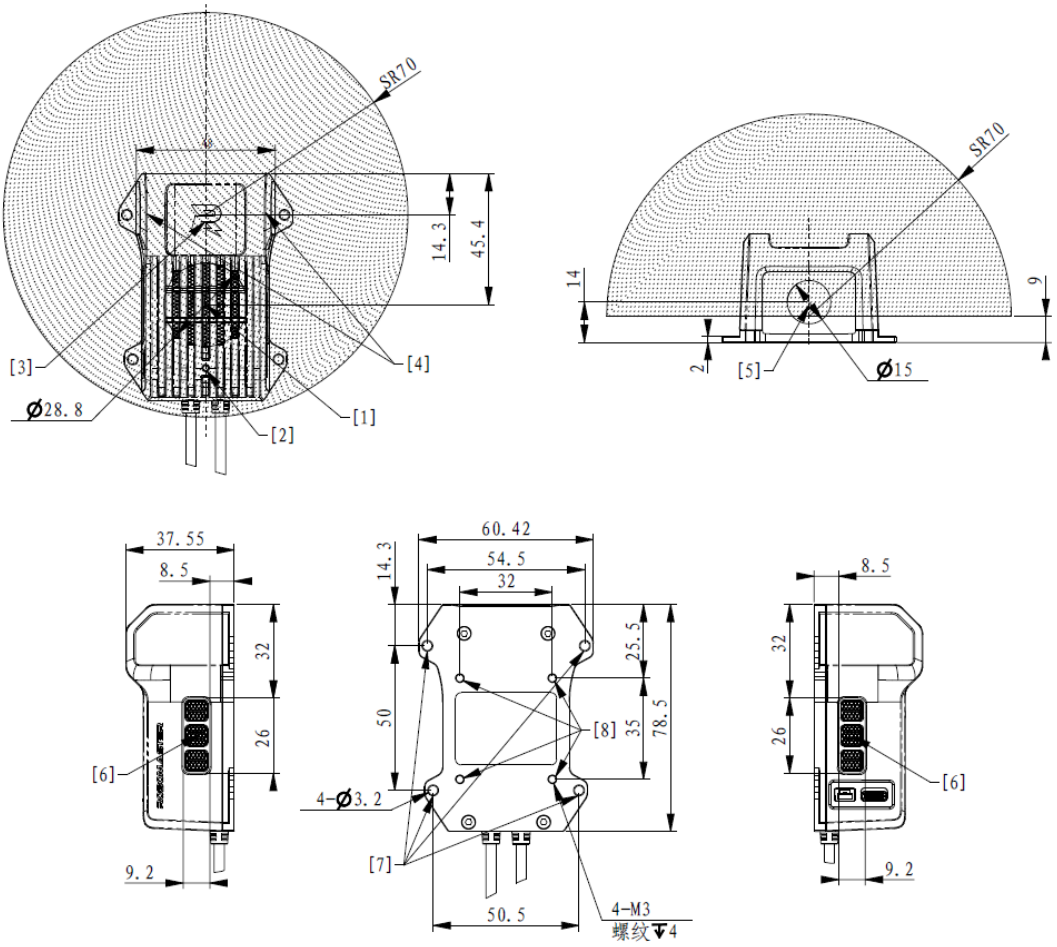


- The effective detection distance of the Field Interaction Module is 100 mm (±5%); the actual distance after installation must be determined by testing. If the effective distance is reduced or the module does not work properly, check the installation. For the specific effective detection distance, see the Field Interaction Module Manual and Power Management Module Manual.
- Because the electromagnetic environment inside a robot is complex, test the effective detection distance while all robot modules are operating normally (for example, with the Supercapacitor, drive motors, and wireless charging coil operating). If the robot has multiple operating conditions, such as capacitor charging/discharging or motor speed changes, also test the distance under each condition.

3.9 Camera Video Transmitter Module (Transmitter) Installation Specifications

- Video-transmission link data is output from the UART serial port of the Camera Video Transmitter Module (Transmitter).
- The UART serial port on the Camera Video Transmitter Module (Transmitter) supports 3.3 V TTL logic. Its UART pins are RX, TX, and GND. Do not connect the positive power supply to the UART port.

Refer to the Transmitter dimensions and mounting interfaces and reserve mounting holes at a suitable position.



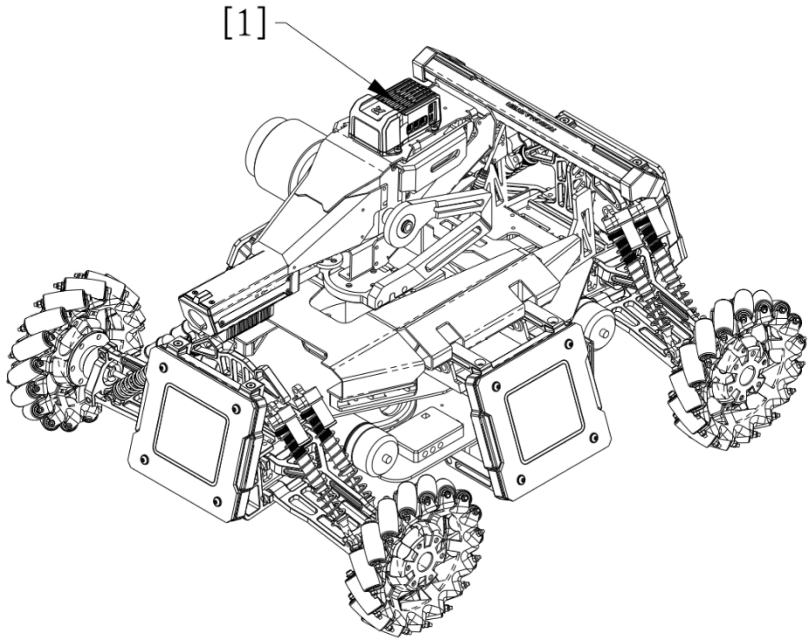
[1] Air inlet	[2] Indicator	[3] Center position	[4] Antenna
[5] Camera	[6] Air outlet	[7] Through mounting hole	[8] Threaded mounting hole

Figure 3-40 Camera Video Transmitter Module (transmitter) diagram

Use the Camera Video Transmitter Module (Transmitter) shown above as the reference. Transmitters shown in other drawings are for reference only.

3.9.1 Installation Steps

1. Use four M3 screws to secure the Transmitter at a suitable position on the robot.



[1] Camera Video Transmitter Module (Transmitter)

Figure 3-41 Camera Video Transmitter Module (transmitter) installation diagram

2. Use an aviation adapter cable to connect the aviation cable interface on the Camera Video Transmitter Module (Transmitter) to the video interface on the Power Management Module.

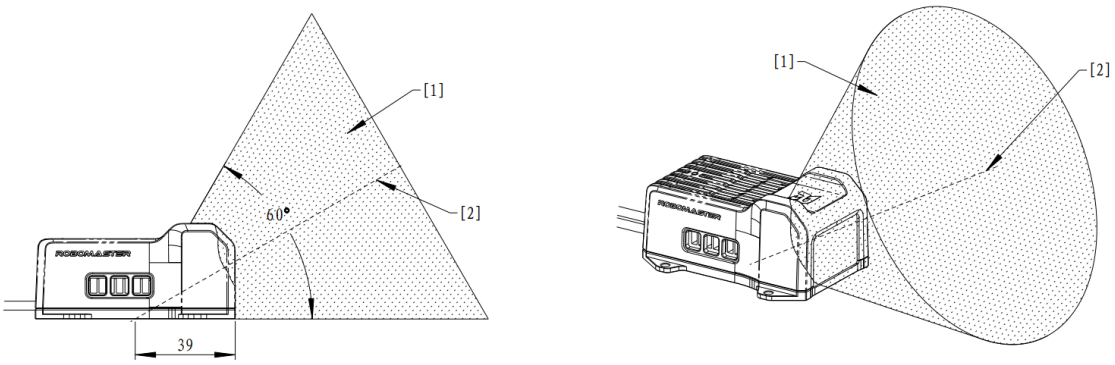
3.9.2 Installation Requirements

Install the Camera Video Transmitter Module (Transmitter) in accordance with these requirements; otherwise, image quality may decline, video-link transmission may become unstable, or the robot may unexpectedly disconnect.

- Do not obstruct the indicator on the Camera Video Transmitter Module (Transmitter).
- With the center of the bottom surface of the Camera Video Transmitter Module (Transmitter) as the vertex, there must be an unobstructed cone directly in front of the camera. Its generatrices are parallel to the bottom surface, it has a 60° cone angle (solid angle $\pi(2 - \sqrt{3})$ sr), and it extends infinitely; see the figure below.

S147 Do not obstruct the air inlet or air outlet of the Camera Video Transmitter Module (Transmitter).

Recommendation: Keep the area within 150 mm of each air inlet and outlet unobstructed.



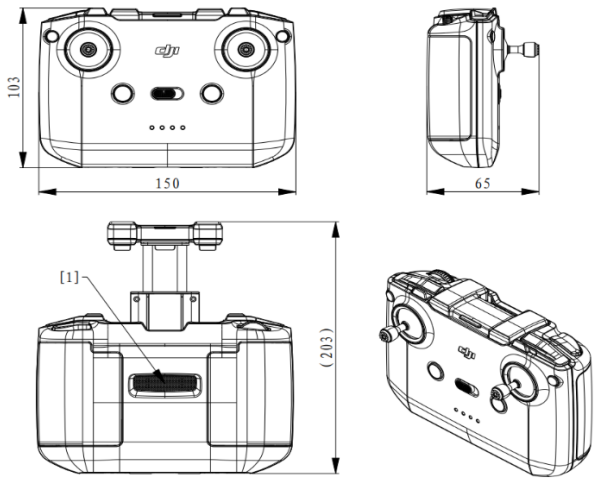
[1] Prohibited-obstruction zone

[2] Zone axis

Figure 3-42 No-obstruction zone of the Camera Video Transmitter Module (transmitter)

- Within the upper hemisphere of 70 mm radius centered on the midpoint of the line connecting the bottom surfaces of the Transmitter antennas, no conductive material, electromagnetic-absorbing material, or other component that may interfere with wireless signals is allowed, as shown in “Figure 3-40 Camera Video Transmitter Module (Transmitter) Diagram.”
- If using the video-transmission link, protect the UART port on the Camera Video Transmitter Module (Transmitter), for example, with foam or a nonmetallic baffle.
- Do not obstruct the interfaces or indicator on the Camera Video Transmitter Module (Transmitter).

3.10 Camera Video Transmitter Module (Receiver) Usage Specifications



[1] Ventilation hole

Figure 3-43 Camera Video Transmitter Module (receiver) diagram

3.10.1 Installation Requirements

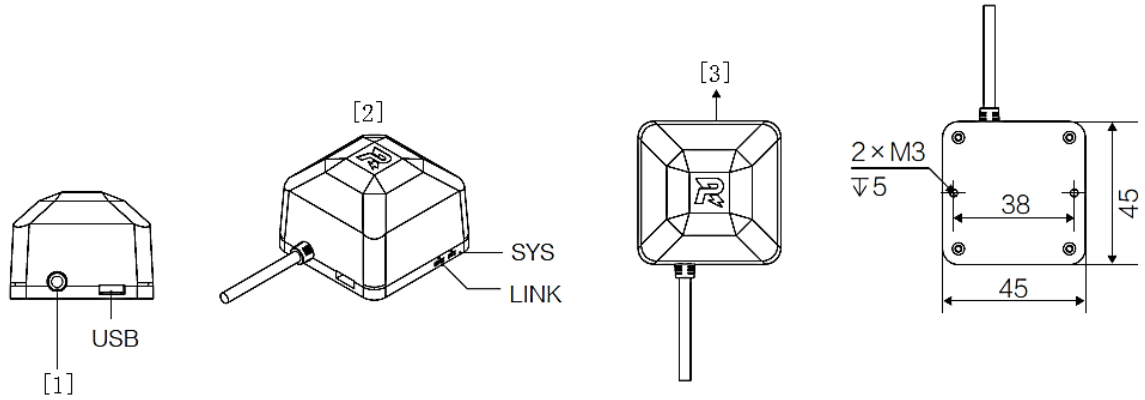
Use the Camera Video Transmitter Module (Receiver) in accordance with the following requirements; otherwise, image quality may decline, remote-control data transmission may stutter, or the module may malfunction.

S148 Avoid metal obstructions near the Camera Video Transmitter Module (Receiver).

S149 Do not obstruct the cooling outlet.

3.11 Positioning Module Installation Specifications

Refer to the Positioning Module dimensions and reserve mounting holes at a specific position on the robot.



[1] Referee System cable [2] Top [3] Front

Figure 3-44 Positioning Module diagram

3.11.1 Installation Steps

1. Use two M3 screws to secure the Positioning Module at the designated position on the robot, as shown below:

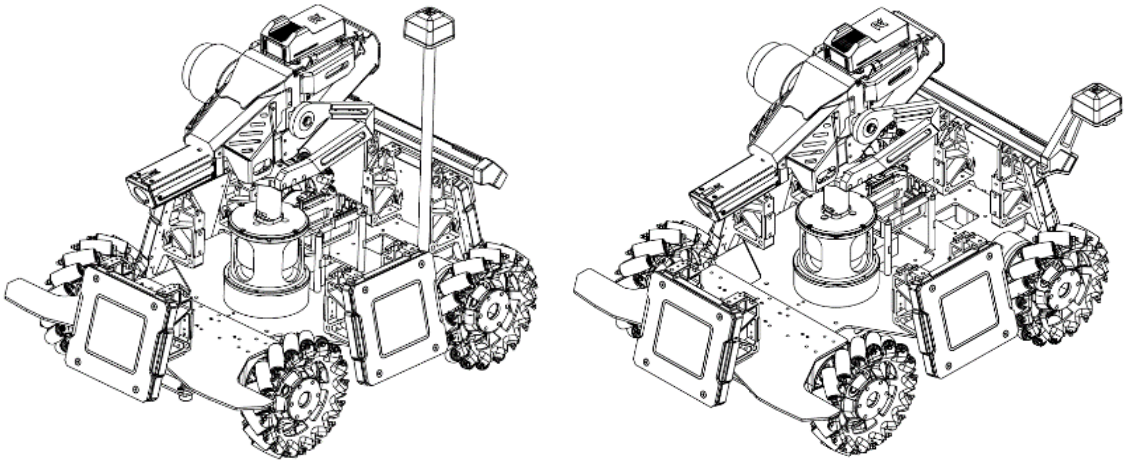


Figure 3-45 Positioning Module installation diagram

2. Use the aviation adapter cable included in the package to connect the Positioning Module to the aviation cable interface with a silver-white metal ring on the Power Management Module.

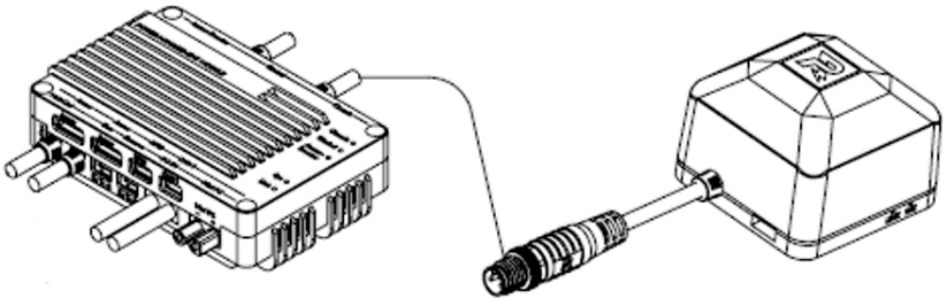


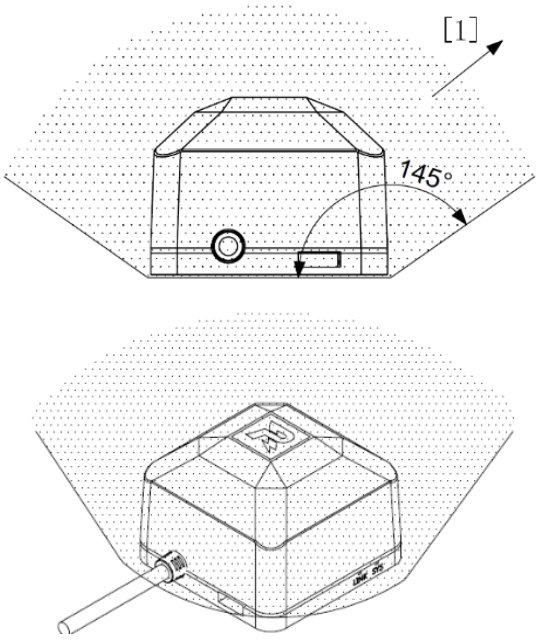
Figure 3-46 Positioning Module wiring diagram

3.11.2 Installation Requirements

Install the Positioning Module in accordance with the following requirements; otherwise, positioning may be abnormal.

S150 The Positioning Module mounting bracket must be rigidly connected to the robot. When a 40 N force is applied perpendicular to any point on the bracket, the plane of the bottom of the Positioning Module must deviate by no more than 10° from horizontal.

S151 Install the Positioning Module level with its top facing upward. Conductive objects must not obstruct the 145° region above it, as shown below. If installed on a mechanism that can actively transform or is part of a mechanical linkage, the Positioning Module must meet the obstruction requirements before, during, and after robot transformation.



[1] Extends infinitely

Figure 3-47 Positioning Module installation diagram

S152 Install the Positioning Module at least 150 mm from the Camera Video Transmitter Module. Measure the shortest straight-line distance between their outer edges.

S153 No magnets, magnetically conductive materials, easily magnetized objects, or components that generate a strong magnetic field may be within 100 mm of the Positioning Module, to avoid interference.

Common sources of magnetic interference include motors, magnetized metal, and magnets.

S154 Do not use magnets, magnetically conductive materials, or adhesive materials to secure the Positioning Module.

- Mounting screws are included with the Positioning Module. Their use is recommended; replacements are available at the venue if the screws are lost.
- If using other screws, they must meet the relevant specifications.

3.12 Supercapacitor Management Module Installation Specifications

The Supercapacitor Management Module measures the capacitance of the Supercapacitor module and its energy during a match. Its expected dimensions are 60 × 30 × 7.5 mm (L × W × H), and heat-shrink tubing is used as its outer protection.

Hardware interfaces: one XT30 male connector, two XT30 female connectors, and one Supercapacitor Management Module communication interface.

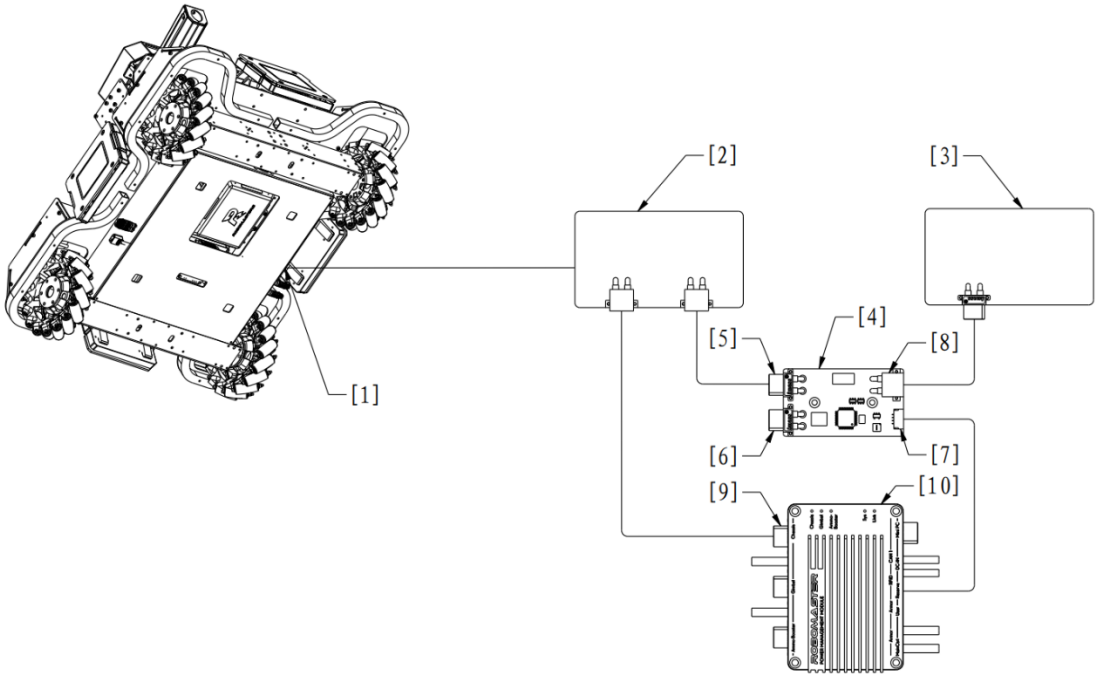
- XT30 female connector model: XT30PW-F.
- XT30 male connector model: XT30PW-M.

3.12.1 Installation Steps

The team-built power-control board adjusts output power from the Power Management Module Chassis interface and input/output power of the Supercapacitor module to meet the rules' power limits.

1. Install the Supercapacitor Management Module between the Supercapacitor module output and the power-control board input.
2. Use cables with XT30 connectors to connect the Supercapacitor module to the Supercapacitor Management Module.
3. Use cables with XT30 connectors to connect the power-control board to the Supercapacitor Management Module.
4. Use a 4-pin cable to connect the communication interface on the Supercapacitor Management Module to the CAN1 interface on the Power Management Module.

The Supercapacitor Management Module wiring is shown below:



[1] Robot chassis power interface

[2] Power-control board

[3] Supercapacitor module

[4] Supercapacitor Management Module

[5] Supercapacitor Management Module interface (output, XT30 female) — connects to the power-control board

[6] Supercapacitor Management Module inspection interface (output, XT30 female) — for inspection only

[7] Supercapacitor Management Module communication interface (CAN, SM04B-GHS-TB) — connects to the Power Management Module

[8] Supercapacitor Management Module interface (input, XT30 male) — connects to the Supercapacitor module

[9] Chassis output interface on the Power Management Module

[10] Power Management Module

Figure 3-48 Supercapacitor Management Module wiring diagram

3.12.2 Installation Requirements

- The communication interface on the Supercapacitor Management Module must connect to CAN1 on the Power Management Module; otherwise, it will not work properly.
- During Pre-Match Inspection, staff will use the Supercapacitor Management Module wiring diagram to check whether the team's Supercapacitor module is installed correctly. After inspection, they will place a pass label on the module and capacitor bank. Teams are advised to design a structure that makes the label easy to see, avoiding frequent robot disassembly during inspection.

S155 The peak and maximum continuous current at the XT30 interfaces on the Supercapacitor Management Module must each not exceed 15 A.

S156 The capacitor voltage must be reduced from fully charged to below 1 V. An abnormal rapid voltage drop or similar condition will fail Pre-Match Inspection.

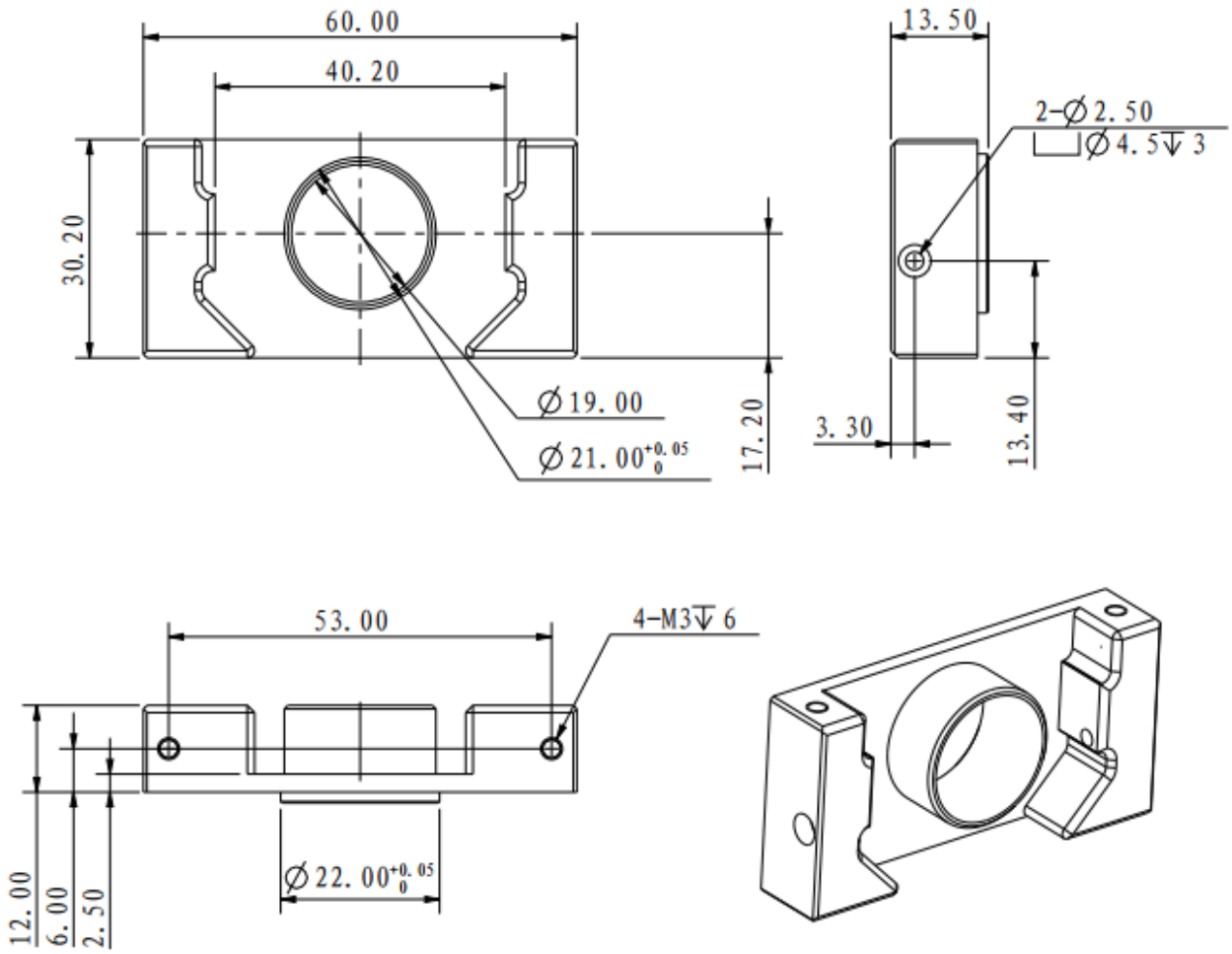
S157 The inspection interface on the Supercapacitor Management Module must have an XT30 female-connector cable at least 100 mm long.

S158 Heavy Robots, Infantry Robots, and Sentry Robots must install a Supercapacitor Management Module. If a robot has no Supercapacitor module, connect the Supercapacitor Management Module to the Power Management Module with a 4-pin cable only.

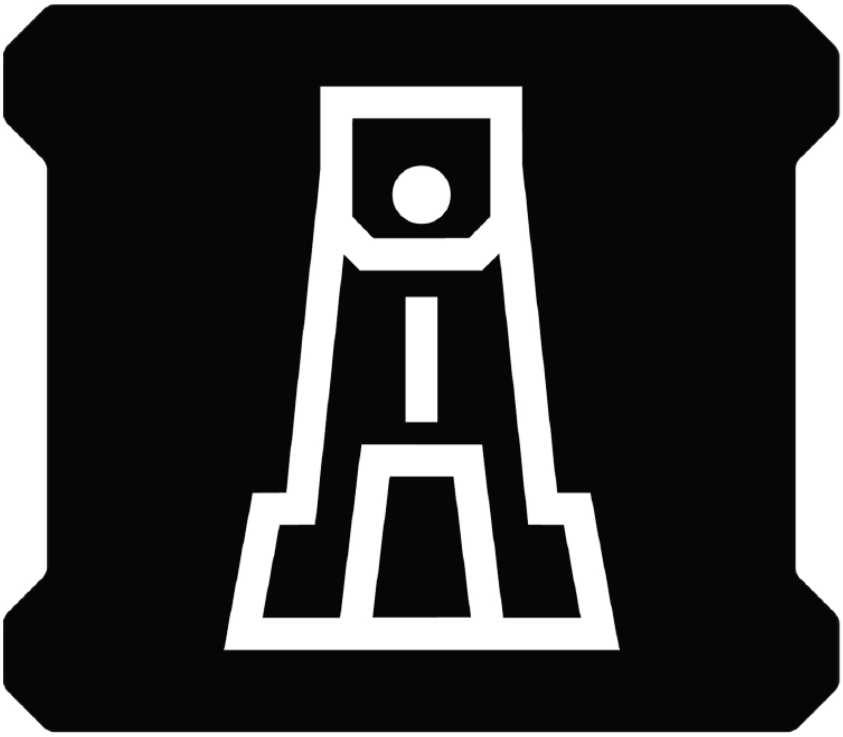
S159 Install the Supercapacitor Management Module where it is easy to access so it can be operated during Pre-Match Inspection.

S160 The Supercapacitor module may use only one power interface and must be connected as shown in “Figure 3-48 Supercapacitor Management Module Wiring Diagram.” Do not connect it to any other robot circuit.

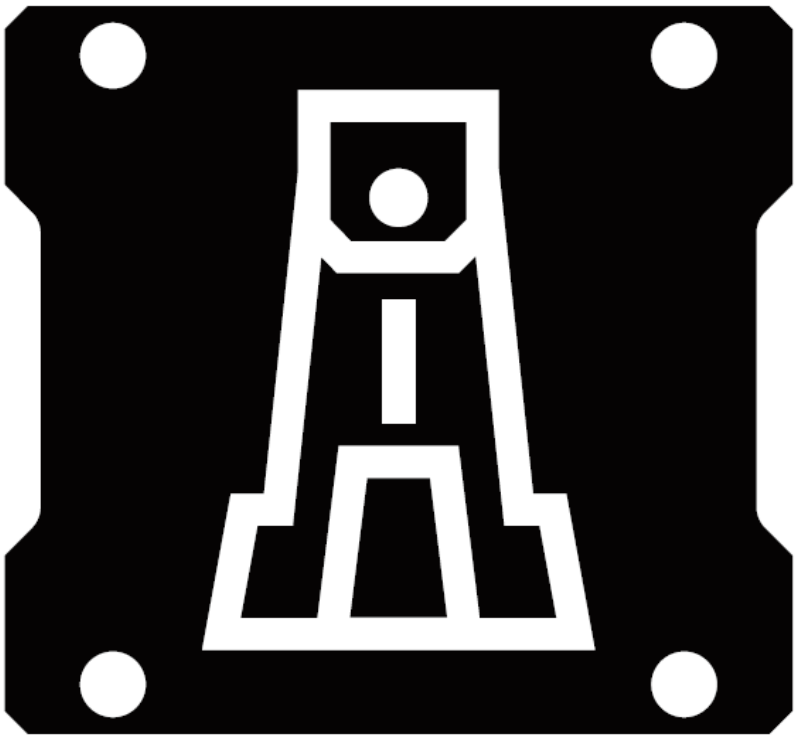
Appendix 1 17mm Speed Monitor Module Adapter Part Engineering Drawing



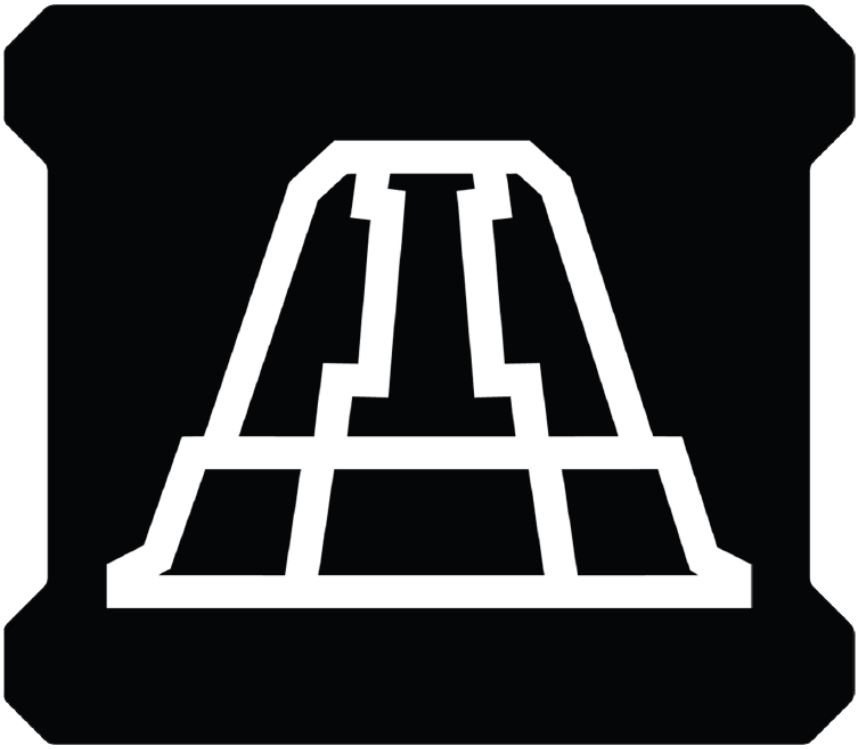
Appendix 2 Reference Drawings



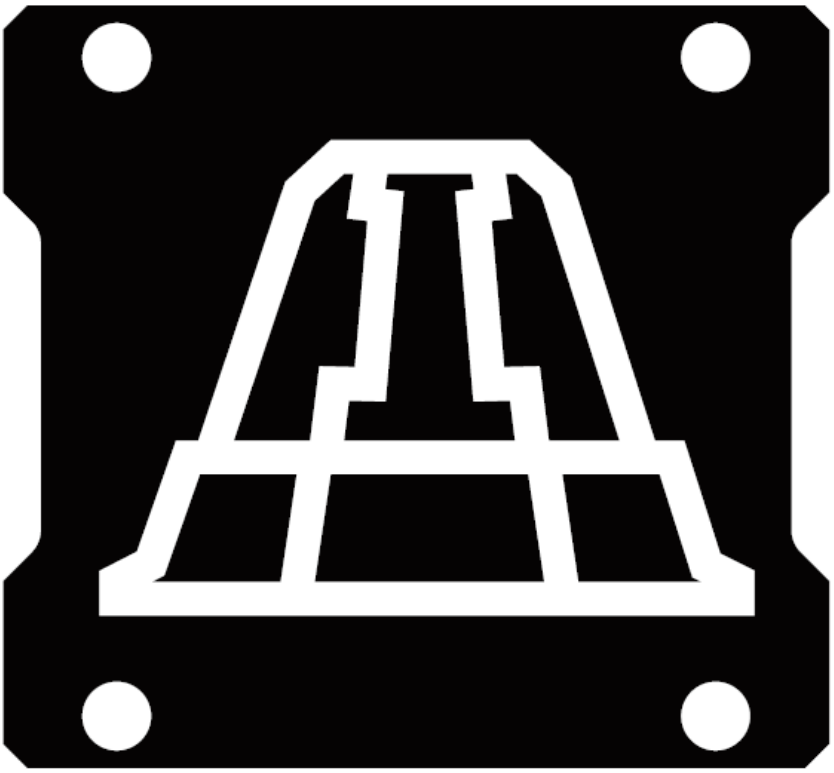
Appendix Figure 1 Outpost Armor Sticker



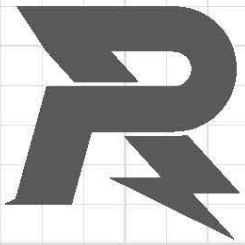
Appendix Figure 2 Outpost Dart Detection Module Sticker



Appendix Figure 3 Base Small Armor Sticker



Appendix Figure 4 Base Dart Detection Module Sticker



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